

Reinforcement Learning Companion Workbook

Exercises for Chapters 1–10

Problems Only

Original companion exercises for Shiyu Zhao's Mathematical Foundations of Reinforcement Learning.

This document contains 305 original exercises, organized chapter by chapter. Chapter 1 contains 35 foundational exercises, and Chapters 2–10 contain 30 exercises each. Across the complete workbook, 213 exercises are numerical, derivational, or structured problem-solving questions and 92 are conceptual or theory questions, giving an overall distribution of approximately 70/30. Within each chapter, the exercises progress from basic understanding to intermediate applications and then to challenging synthesis. Every question is restricted to concepts introduced no later than the chapter in which it appears. Solutions are intentionally omitted from this problem-only edition.

Unless an exercise states otherwise, state and action spaces are finite, rewards are received on transitions, policies and environment models are stationary, and discounted problems use $0 < \gamma < 1$. Show derivations and justify conclusions; unsupported numerical answers are not sufficient.

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Chapter 1: Basic Concepts

This chapter covers agents and environments, states, actions, transition models, policies, rewards, trajectories, discounted returns, episodes, and Markov decision processes. Bellman equations from Chapter 2 are not assumed.

Part A: Conceptual Foundations

Exercise 1.1. Agent and environment

(Conceptual/theory; Basic)

For each system, identify the **agent** and the **environment**:

- a. A robot vacuum cleaning an apartment
- b. A program playing chess
- c. A recommendation system selecting videos
- d. A self-driving car approaching an intersection

Exercise 1.2. State design

(Conceptual/theory; Basic)

A robot vacuum operates inside a house.

Consider these three possible state definitions:

1. The robot's current room
2. The robot's current room and battery level
3. The complete sequence of rooms previously visited

For each definition, explain:

- a. What information it contains
- b. What information it ignores
- c. Whether it would likely be sufficient for making good decisions

Exercise 1.3. Actions versus consequences

(Conceptual/theory; Basic)

A robot selects the action “move right,” but wind pushes it downward.

Identify:

- a. The action
- b. The resulting state transition
- c. Why an action and its consequence are not necessarily the same thing

Exercise 1.4. State space and action space

(Numerical/problem-solving; Basic)

A drone can occupy one of 20 locations and can choose among

{north, south, east, west, hover}.

Determine:

- What is the size of the state space?
- What is the size of the action space?
- How many state–action pairs are possible if every action is available in every state?

Exercise 1.5. State-dependent action spaces

(Numerical/problem-solving; Basic)

An elevator is on the ground floor of a ten-floor building.

Why might its action space differ from the action space on the fifth floor? Write suitable action spaces for both states.

Part B: Grid-World Setup

Use the following 3×3 grid for Exercises 1.6–1.18:

s_1	s_2	s_3
s_4	s_5	s_6
s_7	s_8	s_9

The available actions are

$$A = \{U, R, D, L, S\},$$

where S means stay still.

Rules

- Attempting to cross the boundary leaves the agent in its current state.
- All cells are accessible.
- s_6 is forbidden.
- s_9 is the target.

Part C: State Transitions

Exercise 1.6. Deterministic transitions

(Numerical/problem-solving; Basic)

Find the next state for each state–action pair:

- (s_1, R)
- (s_1, U)
- (s_5, D)
- (s_5, R)
- (s_9, R)
- (s_7, S)

Exercise 1.7. Complete transition row

(Numerical/problem-solving; Basic)

Write the next state for every action from s_2 :

Current state	U	R	D	L	S
s_2	?	?	?	?	?

Exercise 1.8. Transition probability

(Numerical/problem-solving; Basic)

The environment is deterministic. The agent takes R in s_4 .

Write

$$p(s_i \mid s_4, R)$$

for all nine possible next states s_i .

Exercise 1.9. Stochastic wind

(Numerical/problem-solving; Basic)

Suppose that when the agent chooses R in s_1 :

- It moves to s_2 with probability 0.8.
- Wind pushes it to s_4 with probability 0.2.

Write the complete next-state probability distribution

$$p(s' \mid s_1, R).$$

Verify that it is a valid probability distribution.

Exercise 1.10. Two sources of uncertainty

(Conceptual/theory; Basic)

Explain the difference between:

- A stochastic transition model
- A stochastic policy

Give an example in which the environment is deterministic but the policy is stochastic.

Part D: Policies

Exercise 1.11. Valid policy or not?

(Numerical/problem-solving; Basic)

At state s_1 , consider

$$\pi(U \mid s_1) = 0, \quad \pi(R \mid s_1) = 0.4, \quad \pi(D \mid s_1) = 0.4,$$

$$\pi(L \mid s_1) = 0, \quad \pi(S \mid s_1) = 0.2.$$

Is this a valid policy distribution? Explain.

Exercise 1.12. Invalid policies

(Numerical/problem-solving; Basic)

Determine what is wrong with each proposed policy:

a.

$$\pi(R \mid s) = 0.7, \quad \pi(D \mid s) = 0.5,$$

with all other probabilities zero.

b.

$$\pi(R \mid s) = 1.2, \quad \pi(D \mid s) = -0.2,$$

with all other probabilities zero.

Exercise 1.13. Deterministic or stochastic

(Conceptual/theory; Basic)

Classify each policy:

- At every state, exactly one action has probability one.
- At s_1 , the agent chooses R or D , each with probability 0.5.
- At s_5 , the agent chooses S with probability one.
- Every action is selected with probability 0.2.

Exercise 1.14. Construct a policy

(Numerical/problem-solving; Intermediate)

Design a deterministic policy that moves the agent from every non-target state toward s_9 , while avoiding s_6 .

Represent the policy as a table:

State	Selected action
s_1	
s_2	
$\vdots$	
s_9	

There may be more than one reasonable answer.

Exercise 1.15. Policy-induced trajectory

(Numerical/problem-solving; Intermediate)

Suppose the deterministic policy is

$$\pi(s_1) = R, \quad \pi(s_2) = D, \quad \pi(s_5) = D, \quad \pi(s_8) = R, \quad \pi(s_9) = S.$$

Starting from s_1 , write the first seven states in the resulting trajectory.

Exercise 1.16. Policy and transition uncertainty together

(Numerical/problem-solving; Intermediate)

At s_1 , a policy chooses

$$\pi(R \mid s_1) = 0.6, \quad \pi(D \mid s_1) = 0.4.$$

The environment behaves as follows:

- R reaches s_2 with probability 0.9 and s_4 with probability 0.1.
- D always reaches s_4 .

Calculate

$$P(S_{t+1} = s_2 \mid S_t = s_1)$$

and

$$P(S_{t+1} = s_4 \mid S_t = s_1).$$

Part E: Rewards

For Exercises 1.17–1.20, use

$$\begin{aligned}r_{\text{boundary}} &= -1, & r_{\text{forbidden}} &= -1, \\r_{\text{target}} &= +1, & r_{\text{other}} &= 0.\end{aligned}$$

The reward is determined by what happens after executing the action.

Exercise 1.17. Immediate rewards

(Numerical/problem-solving; Basic)

Determine the immediate reward for:

- Taking U at s_1
- Taking R at s_1
- Taking R at s_5
- Taking D at s_6
- Taking S at s_9
- Taking R at s_9

Exercise 1.18. Immediate reward is not enough

(Numerical/problem-solving; Intermediate)

At a state s , two actions are available:

- Action a_1 gives an immediate reward of $+2$, followed by rewards $-5, -5$.
- Action a_2 gives an immediate reward of 0 , followed by rewards $+2, +2$.

Without discounting:

- Calculate the return for each action.
- Which action appears better?
- Why would choosing only the action with the largest immediate reward fail?

Exercise 1.19. Reward-design failure

(Conceptual/theory; Intermediate)

A warehouse robot receives:

- $+100$ for delivering a package.
- No penalty for damaging equipment.
- No penalty for taking excessive time.

Describe two undesirable behaviours that could emerge from this reward design.

Exercise 1.20. Alternative reward schemes

(Numerical/problem-solving; Intermediate)

Design two different reward schemes for the grid world:

- One that strongly prioritizes safety
- One that prioritizes reaching the target quickly

Explain how the resulting behaviour might differ.

Part F: Trajectories and Returns

Exercise 1.21. Identify a trajectory

(Numerical/problem-solving; Basic)

Consider

$$s_1 \xrightarrow{R; 0} s_2 \xrightarrow{D; 0} s_5 \xrightarrow{R; -1} s_6 \xrightarrow{D; +1} s_9.$$

List separately:

- The state sequence
- The action sequence
- The reward sequence
- The undiscounted return

Exercise 1.22. Compare two trajectories

(Numerical/problem-solving; Basic)

Trajectory A produces rewards

$$0, 0, 0, +1.$$

Trajectory B produces

$$0, -1, 0, +1.$$

Calculate their undiscounted returns and determine which trajectory is preferable under this criterion.

Exercise 1.23. Discounted return

(Numerical/problem-solving; Basic)

For rewards

$$R_1 = 2, \quad R_2 = -1, \quad R_3 = 4$$

and discount rate

$$\gamma = 0.5,$$

calculate

$$G_0 = R_1 + \gamma R_2 + \gamma^2 R_3.$$

Exercise 1.24. Effect of the discount rate

(Numerical/problem-solving; Intermediate)

A trajectory produces rewards

$$0, 0, 0, 10.$$

Calculate its discounted return using:

- $\gamma = 0$
- $\gamma = 0.5$
- $\gamma = 0.9$
- $\gamma = 1$

Explain what the results reveal about short-sighted and far-sighted behaviour.

Exercise 1.25. Earlier versus larger reward

(Numerical/problem-solving; Intermediate)

An agent must choose between:

- Policy A: rewards 0, 4, 0, 0
- Policy B: rewards 0, 0, 0, 6

For each discount rate, determine which policy has the greater return:

- a. $\gamma = 0.5$
- b. $\gamma = 0.9$

Show the complete calculations.

Exercise 1.26. Infinite trajectory

(Numerical/problem-solving; Intermediate)

After reaching s_9 , the agent remains there and receives +1 forever.

Calculate the return beginning at s_9 when:

- a. $\gamma = 0.5$
- b. $\gamma = 0.9$
- c. $\gamma = 0.99$

Use

$$1 + \gamma + \gamma^2 + \dots = \frac{1}{1 - \gamma}.$$

What happens as γ approaches one?

Exercise 1.27. Delayed infinite reward

(Numerical/problem-solving; Intermediate)

The agent receives rewards

$$0, 0, 0, 1, 1, 1, \dots$$

Write the discounted return as a geometric series and simplify it into a closed-form expression involving γ .

Part G: Episodes and Task Types

Exercise 1.28. Episodic or continuing?

(Conceptual/theory; Basic)

Classify each task:

- a. Playing one game of chess
- b. Temperature control in a data centre
- c. Completing one maze
- d. Managing an investment portfolio indefinitely
- e. A robot delivering one package

- f. A recommendation platform that operates continuously

Explain each classification.

Exercise 1.29. Terminal and absorbing states

(Conceptual/theory; Intermediate)

Explain the difference between:

- a. A terminal state
- b. An absorbing state
- c. A normal state in which the “stay” action is available

Can a terminal state be mathematically converted into an absorbing state?

Part H: Markov Decision Processes

Exercise 1.30. MDP components

(Numerical/problem-solving; Intermediate)

For a traffic-light controller, propose:

- a. A state space
- b. An action space
- c. A transition model
- d. A reward function
- e. A policy

Your state definition should include enough information for decision-making.

Exercise 1.31. Testing the Markov property

(Conceptual/theory; Intermediate)

A robot’s state is defined only as its current location.

Its battery level affects whether it can successfully move, but battery level is not included in the state.

Does the resulting state representation satisfy the Markov property? Explain precisely.

Exercise 1.32. Repairing a non-Markov state

(Numerical/problem-solving; Intermediate)

For Exercise 1.31, propose a better state definition that makes the process closer to Markov.

Exercise 1.33. Medical-treatment example

(Conceptual/theory; Intermediate)

A treatment system defines the patient’s state only as “sick” or “healthy.” Treatment outcomes also depend strongly on:

- Age
- Previous medication
- Disease severity
- Allergies

Explain why the two-state representation is inadequate from an MDP perspective.

Exercise 1.34. MDP versus Markov chain

(Conceptual/theory; Intermediate)

Explain what happens when a policy is fixed in an MDP.

Why can the resulting system be viewed as a Markov process or Markov chain?

Part I: Integrated Challenge

Exercise 1.35. Build a complete MDP

(Numerical/problem-solving; Challenging)

Create your own 4×4 grid world containing:

- One starting state
- One target
- Two forbidden cells
- At least one stochastic transition
- Five actions
- Boundary collisions
- A complete reward specification
- One deterministic policy
- One stochastic policy

Then provide:

- a. The state space
- b. The action space
- c. Three example transition distributions
- d. Three example policy distributions
- e. One complete episode
- f. Its undiscounted return
- g. Its discounted return for $\gamma = 0.9$
- h. An explanation of whether your chosen state representation is Markov

Chapter 2: State Values and the Bellman Equation

This chapter covers returns and state values, Bellman equations, policy-induced transition models, matrix and iterative policy evaluation, and action values. Optimization concepts introduced in Chapter 3 are not assumed.

Notation

At time t , the state, action, next reward, and return are denoted by

$$S_t, \quad A_t, \quad R_{t+1}, \quad G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}, \quad 0 < \gamma < 1.$$

For a fixed stationary policy π ,

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s], \quad q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a].$$

Unless stated otherwise, rewards are received on transitions, state and action spaces are finite, and the policy and environment model are stationary.

Part A: Returns and State Values

Exercise 2.1. A state value as an average of returns

(Numerical/derivation; Basic)

Starting from state s , a stochastic policy produces one of two complete three-reward trajectories:

trajectory	probability	$(R_{t+1}, R_{t+2}, R_{t+3})$
A	0.65	$(2, 0, 2)$
B	0.35	$(-1, 4, 4)$

The episode terminates after the third reward and $\gamma = 0.8$.

- Calculate the return on each trajectory.
- Calculate $v_{\pi}(s)$.
- If the policy were changed to choose trajectory A with probability one, what would the new value of s be?
- Explain why the answer to part (b) need not equal either realized return.

Exercise 2.2. Return, state value, and randomness

(Conceptual/theory; Basic)

Decide whether each statement is always true. If it is false, correct it.

- “The return G_t and the state value $v_{\pi}(s)$ are the same object.”
- “If both the policy and the environment are deterministic, the value of a state equals the return from that state.”
- “If $v_{\pi}(s) = 7$, every trajectory starting from s has return 7.”
- “Changing the policy can change $v_{\pi}(s)$ even when the environment model is unchanged.”

Exercise 2.3. A random first reward followed by a continuing reward*(Numerical/derivation; Basic)*

From state s , the next reward is $+3$ with probability 0.4 and -1 with probability 0.6 . In either case, the next state is h . Starting from h , every subsequent transition gives reward $+2$ forever. Let $\gamma = 0.75$.

- Find $v_\pi(h)$.
- Find the mean immediate reward on leaving s .
- Use a one-step decomposition to find $v_\pi(s)$.
- Write the two possible returns from s and verify that their probability-weighted average agrees with part (c).

Exercise 2.4. Policy comparison depends on the starting state*(Numerical/derivation; Intermediate)*

For two policies π_A and π_B , the possible discounted reward sequences from states x and y are deterministic:

	x	y
π_A	$(1, 1, 1, \dots)$	$(-2, 3, 3, 3, \dots)$
π_B	$(0, 2, 2, 2, \dots)$	$(4, 0, 0, \dots)$

Let $\gamma = 0.8$.

- Compute all four state values.
- Does either policy have a greater value at both states?
- Suppose the initial state is x with probability p and y with probability $1 - p$. Compute the mean initial value $J_A(p)$ and $J_B(p)$.
- For which values of p is π_A preferred under this initial distribution? What happens at $p = 0.7$?

Exercise 2.5. Why the time index disappears*(Conceptual/theory; Intermediate)*

Chapter 2 writes $v_\pi(s)$, not $v_{\pi,t}(s)$.

- Explain which stationarity assumptions make the value independent of the calendar time t .
- What if the reward for the same transition doubles after time $t = 100$, but time is not part of the state?
- Give two mathematically sound ways to repair the notation or the state representation in part (b).

Part B: Building the Bellman Equation**Exercise 2.6. A one-state Bellman equation***(Numerical/derivation; Basic)*

There is one state s , and both actions return to s . Action a gives reward 2 , action b gives reward -1 , and

$$\pi(a \mid s) = 0.3, \quad \pi(b \mid s) = 0.7, \quad \gamma = 0.6.$$

- a. Write the Bellman equation for $v_\pi(s)$.
- b. Solve for $v_\pi(s)$.
- c. Compute $q_\pi(s, a)$ and $q_\pi(s, b)$.
- d. Verify $v_\pi(s) = \sum_c \pi(c | s) q_\pi(s, c)$.

Exercise 2.7. Bootstrapping around a two-state cycle

(Numerical/derivation; Basic)

Under a deterministic policy, state x transitions to y with reward 3, and y transitions to x with reward -1 . Let $\gamma = 0.5$.

- a. Write the two Bellman equations.
- b. Solve them by substitution.
- c. Starting from x , also evaluate the infinite reward series directly by grouping it into two-step blocks.
- d. Explain why the direct return and the Bellman solution agree.

Exercise 2.8. Stochastic rewards and stochastic transitions

(Numerical/derivation; Intermediate)

A fixed policy induces the following model:

- From x , the immediate reward is $+2$ with probability 0.75 and -2 with probability 0.25. The next state is x with probability 0.4 and y with probability 0.6.
- From y , the immediate reward is always -1 . The next state is x or y , each with probability 0.5.

Use $\gamma = 0.8$. The reward and next-state draws are conditionally independent given the current state.

- a. Compute the two expected immediate rewards.
- b. Write the two Bellman equations.
- c. Solve for $v_\pi(x)$ and $v_\pi(y)$, showing the elimination steps.
- d. Explain how x can have positive value even though it can lead to the negative-value state y .

Exercise 2.9. Where the Markov property enters

(Conceptual/theory; Intermediate)

In deriving the Bellman equation, one replaces

$$\mathbb{E}[G_{t+1} | S_t = s, S_{t+1} = s'] \quad \text{by} \quad \mathbb{E}[G_{t+1} | S_{t+1} = s'].$$

- a. Explain precisely why this replacement is justified for a Markov state and a policy that depends only on the current state.
- b. Give an example with a hidden persistent “mode” in which the replacement fails.
- c. Explain how augmenting the state can restore the derivation.

Exercise 2.10. Equivalent Bellman expressions from a joint model*(Numerical/derivation; Intermediate)*

At state s , the policy satisfies

$$\pi(a \mid s) = 0.4, \quad \pi(b \mid s) = 0.6.$$

The nonzero joint probabilities $p(s', r \mid s, c)$ are

$$\begin{array}{l|l} c = a & (x, 2) : 0.3 \quad (y, -1) : 0.2 \quad (y, 3) : 0.5 \\ c = b & (x, 0) : 0.6 \quad (y, 1) : 0.4 \end{array}$$

and $\gamma = 0.5$. For this calculation, suppose the next-state values are $v_\pi(x) = 4$ and $v_\pi(y) = -2$.

- Derive $p(s' \mid s, c)$ and $p(r \mid s, c)$ for each action.
- Compute the Bellman right-hand side by first finding, for each action, the mean immediate reward and mean next-state value.
- Compute the same quantity directly from

$$\sum_c \pi(c \mid s) \sum_{s', r} p(s', r \mid s, c) [r + \gamma v_\pi(s')].$$

- Verify that the two forms agree.

Part C: Policy-Induced Models and Matrix Form**Exercise 2.11. Constructing $\mathbf{r}_\pi$ and P_π** *(Numerical/derivation; Intermediate)*

There are two states x, y and two actions a, b . The model is:

(s, c)	mean immediate reward	$p(\cdot \mid s, c)$
(x, a)	2	(0.75, 0.25)
(x, b)	-1	(0, 1)
(y, a)	0	(1, 0)
(y, b)	3	(0.2, 0.8)

where each probability row is ordered as (x, y) . The policy is

$$\pi(a \mid x) = 0.4, \quad \pi(b \mid x) = 0.6, \quad \pi(a \mid y) = 0.7, \quad \pi(b \mid y) = 0.3.$$

- Construct $\mathbf{r}_\pi$ in the state order (x, y) .
- Construct P_π and verify that it is a stochastic matrix.
- Write $\mathbf{v}_\pi = \mathbf{r}_\pi + \gamma P_\pi \mathbf{v}_\pi$ componentwise.
- With $\gamma = 0.9$ and initial guess $\mathbf{v}_0 = (1, -2)^T$, perform one iterative Bellman update.

Exercise 2.12. Solving a three-state Bellman system*(Numerical/derivation; Intermediate)*

For a fixed policy, let

$$P_\pi = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{r}_\pi = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \gamma = \frac{1}{2}.$$

- Write the three scalar Bellman equations.
- Solve the equations in an order that exploits their structure.
- Verify the solution by substituting it into $\mathbf{v}_\pi = \mathbf{r}_\pi + \gamma P_\pi \mathbf{v}_\pi$.

Exercise 2.13. Rows, columns, and a common matrix mistake*(Conceptual/theory; Intermediate)*

Under the column-vector convention of Chapter 2, $[P_\pi]_{ij} = p_\pi(s_j | s_i)$.

- Explain why every row, rather than every column, must sum to one.
- Explain the identity $P_\pi \mathbf{1} = \mathbf{1}$.
- A student writes $\mathbf{v}_\pi = \mathbf{r}_\pi + \gamma P_\pi^T \mathbf{v}_\pi$ without changing any definitions. What is wrong?
- Can a valid row-stochastic matrix also have every column sum to one? If so, does that make the transpose error harmless in general?

Exercise 2.14. Closed-form policy evaluation*(Numerical/derivation; Intermediate)*

Let

$$P_\pi = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}, \quad \mathbf{r}_\pi = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \gamma = 0.5.$$

- Form $I - \gamma P_\pi$.
- Calculate its determinant and inverse.
- Use $\mathbf{v}_\pi = (I - \gamma P_\pi)^{-1} \mathbf{r}_\pi$ to find both state values.
- Check each value in the original Bellman equation.

Exercise 2.15. The inverse as discounted state occupancy*(Numerical/derivation; Challenging)*

Consider the alternating two-state chain

$$P_\pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \gamma = \frac{1}{2}.$$

- Show that $P_\pi^2 = I$.
- Starting from

$$(I - \gamma P_\pi)^{-1} = I + \gamma P_\pi + \gamma^2 P_\pi^2 + \cdots,$$

group the even and odd powers to find the inverse without using the 2×2 inverse formula.

- c. If $\mathbf{r}_\pi = (3, 0)^T$, compute $\mathbf{v}_\pi$.
- d. Interpret entry (i, j) of the inverse as a discounted count of visits to s_j when starting from s_i , including the state at step zero. Use this interpretation to explain both values in part (c).

Part D: Iterative Policy Evaluation

Exercise 2.16. Bellman iterates as truncated returns

(Numerical/derivation; Intermediate)

Let

$$P_\pi = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{r}_\pi = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \gamma = \frac{1}{2},$$

and start iterative policy evaluation from $\mathbf{v}^{(0)} = \mathbf{0}$:

$$\mathbf{v}^{(k+1)} = \mathbf{r}_\pi + \gamma P_\pi \mathbf{v}^{(k)}.$$

- a. Compute $\mathbf{v}^{(1)}, \mathbf{v}^{(2)}$, and $\mathbf{v}^{(3)}$.
- b. Solve for the limiting vector $\mathbf{v}_\pi$.
- c. Explain why, with zero initialization, $\mathbf{v}^{(k)}$ contains the expected discounted sum of the first k rewards.
- d. Use part (c) to explain the pattern of the iterates.

Exercise 2.17. Exact error from a nonzero initialization

(Numerical/derivation; Intermediate)

A one-state process has immediate reward 4, a deterministic self-transition, and $\gamma = 0.75$. Iterative policy evaluation starts from $v^{(0)} = -8$.

- a. Find the exact fixed point v_π .
- b. Compute $v^{(1)}, v^{(2)}, v^{(3)}$.
- c. Let $\delta_k = v^{(k)} - v_\pi$. Derive a closed expression for δ_k and hence for $v^{(k)}$.
- d. Find the smallest integer k for which $|v^{(k)} - v_\pi| < 0.5$.

Exercise 2.18. Why bootstrapping is not circular reasoning

(Conceptual/theory; Intermediate)

A learner objects to

$$v_\pi(s) = r_\pi(s) + \gamma \sum_{s'} p_\pi(s' | s) v_\pi(s')$$

because “an unknown value is being defined using unknown values.”

- a. Explain why the Bellman equation is a simultaneous linear system or fixed-point condition, not an instruction to know one unknown in advance.
- b. Explain how both the closed-form and iterative solutions resolve the apparent circularity.
- c. Give a simple example showing why the condition $\gamma < 1$ matters for a continuing reward loop.

Exercise 2.19. A maximum-error bound*(Numerical/derivation; Challenging)*

Let $\mathbf{v}^{(k+1)} = \mathbf{r}_\pi + \gamma P_\pi \mathbf{v}^{(k)}$, and let $\boldsymbol{\delta}_k = \mathbf{v}^{(k)} - \mathbf{v}_\pi$. Define

$$M_k = \max_i |\delta_k(i)|.$$

- Starting from $\boldsymbol{\delta}_{k+1} = \gamma P_\pi \boldsymbol{\delta}_k$, prove that $M_{k+1} \leq \gamma M_k$. Use only nonnegativity and the row-sum property of P_π .
- Deduce $M_k \leq \gamma^k M_0$.
- If $\gamma = 0.9$ and $M_0 \leq 12$, how many iterations are sufficient to guarantee $M_k \leq 0.1$?
- Is the bound necessarily exact? Explain.

Exercise 2.20. Larger rewards under the same dynamics*(Numerical/derivation; Challenging)*

For a fixed transition matrix and $\gamma = 0.5$, suppose

$$(I - \gamma P_\pi)^{-1} = \begin{bmatrix} \frac{4}{3} & \frac{2}{3} \\ 0 & 2 \end{bmatrix}.$$

Two reward vectors are

$$\mathbf{r}^A = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{r}^B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

- Compute $\mathbf{v}^A$ and $\mathbf{v}^B$.
- Verify directly that $\mathbf{v}^A - \mathbf{v}^B \geq \mathbf{0}$.
- Give the general argument: if $\mathbf{r}^A \geq \mathbf{r}^B$ elementwise under the same P_π , why must $\mathbf{v}^A \geq \mathbf{v}^B$?
- Explain why the “same dynamics” qualification is essential to this argument.

Exercise 2.21. What if the discount factor is at an edge case?*(Conceptual/theory; Challenging)*

Although Chapter 2 assumes $0 < \gamma < 1$, consider the limiting cases.

- What does the Bellman equation become when $\gamma = 0$?
- For one state with reward +1 and a self-transition, what goes wrong at $\gamma = 1$?
- For one state with reward 0 and a self-transition, the equation at $\gamma = 1$ becomes $v = v$. Why does that equation alone fail to identify the actual return?
- Summarize what $0 < \gamma < 1$ buys us for continuing finite-state policy evaluation.

Part E: From State Values to Action Values**Exercise 2.22. Values of selected and unselected actions***(Numerical/derivation; Intermediate)*

At state x , actions a, b, c have the following one-step models:

- a : reward -1 , then x with probability one;

- b : reward 2, then y with probability 0.75 and x with probability 0.25;
- c : reward 4 with probability 0.25 and 0 with probability 0.75, then y with probability one.

Let $\gamma = 0.5$, $v_\pi(y) = 2$, and

$$\pi(a \mid x) = 0.5, \quad \pi(b \mid x) = 0.5, \quad \pi(c \mid x) = 0.$$

- Express $q_\pi(x, a)$ and $q_\pi(x, b)$ in terms of $v_\pi(x)$.
- Use $v_\pi(x) = \sum_d \pi(d \mid x) q_\pi(x, d)$ to solve for $v_\pi(x)$.
- Compute all three action values, including $q_\pi(x, c)$.
- Verify the weighted-average identity for $v_\pi(x)$.

Exercise 2.23. Why a zero-probability action still has value

(Conceptual/theory; Intermediate)

Suppose $\pi(a \mid s) = 0$.

- Explain the conditioning experiment represented by $q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$.
- Why is it incorrect to set $q_\pi(s, a) = 0$ merely because $\pi(a \mid s) = 0$?
- Why does this action value not affect $v_\pi(s)$ under the current policy?
- Distinguish “the action value exists in the model” from “the action value is easy to estimate from data generated by π .”

Exercise 2.24. Recovering a policy probability from action values

(Numerical/derivation; Intermediate)

At a state s ,

$$q_\pi(s, a) = 6, \quad q_\pi(s, b) = 1, \quad q_\pi(s, c) = -2.$$

The policy has the form

$$\pi(a \mid s) = \theta, \quad \pi(b \mid s) = 0.8 - \theta, \quad \pi(c \mid s) = 0.2.$$

- Determine the range of θ for which this is a valid policy.
- Express $v_\pi(s)$ as a function of θ .
- If $v_\pi(s) = 2.9$, find every action probability.
- Show the numerical contribution of each action to the weighted average and verify that the contributions sum to 2.9.

Exercise 2.25. Solving state and action values together

(Numerical/derivation; Challenging)

There are two states x, y , two actions a, b , and $\gamma = 0.5$. Transitions and rewards are deterministic:

	a	b
x	reward 2, next y	reward 0, next x
y	reward -1 , next x	reward 3, next y

At either state, the fixed policy selects a and b with probability 0.5 each.

- Express all four action values in terms of $v_\pi(x)$ and $v_\pi(y)$.
- Average the action values under the policy to obtain two equations for the state values.
- Solve for both state values.
- Compute all four action values and verify both weighted averages.

Exercise 2.26. The matrix action-value Bellman equation

(Numerical/derivation; Challenging)

Use the model and policy from Exercise 2.25. Order state-action pairs as

$$(x, a), (x, b), (y, a), (y, b),$$

and order states as x, y .

- Construct the 4×2 matrix P with $[P]_{(s,c),s'} = p(s' | s, c)$.
- Construct the 2×4 matrix Π with $[\Pi]_{s',(s',c')} = \pi(c' | s')$.
- Compute $P\Pi$ and explain what one of its rows represents.
- Form the immediate-reward vector $\tilde{\mathbf{r}}$.
- Using your action-value vector from Exercise 2.25, verify every row of

$$\mathbf{q}_\pi = \tilde{\mathbf{r}} + \gamma P\Pi \mathbf{q}_\pi.$$

Exercise 2.27. How the future policy enters an action value

(Conceptual/theory; Challenging)

Two policies π and μ are compared at the same state-action pair (s, a) . In both cases, the first action is forced to be a .

- Can $q_\pi(s, a)$ and $q_\mu(s, a)$ differ? Explain exactly where the policy enters after the first action.
- Give a condition under which the two action values must be equal.
- What happens to policy dependence when $\gamma = 0$?
- Explain why the phrase “the value of action a ” is incomplete unless the state and continuation policy are understood.

Part F: Integrated Policy-Evaluation Challenges

Exercise 2.28. Evaluating a parameterized stochastic policy

(Numerical/derivation; Challenging)

There are two states x, y and $\gamma = 0.5$. At y , the policy takes a single action that gives reward 0 and moves to x . At x :

- action a gives reward 1 and moves to y ;
- action b gives reward 4, then moves to x or y with probability 0.5 each.

The policy selects a at x with probability θ and b with probability $1 - \theta$, where $0 \leq \theta \leq 1$.

- Write the Bellman equations for $v_\theta(x)$ and $v_\theta(y)$.

- b. Derive the closed form

$$v_{\theta}(x) = \frac{8(4 - 3\theta)}{5 + \theta},$$

showing every algebraic step, and find $v_{\theta}(y)$.

- c. Evaluate both values at $\theta = \frac{1}{4}$ and $\theta = \frac{3}{4}$.
- d. Suppose the initial state is uniformly distributed over x, y . Compare the mean initial values for these two choices of θ .
- e. For which θ is $v_{\theta}(x) \geq 3$?

Exercise 2.29. Different policies with identical state values

(Conceptual/theory; Challenging)

- a. Construct a one-state continuing example with two genuinely different policies that have the same state value.
- b. Verify the equality with a Bellman equation.
- c. State a general reason why different policies can induce the same full state-value vector.
- d. Does equality of value vectors imply equality of policies? Does a fixed policy and fixed model have more than one value vector when $0 < \gamma < 1$?

Exercise 2.30. Four-state synthesis

(Numerical/derivation; Challenging)

For a fixed policy, the induced model in the state order (s_1, s_2, s_3, s_4) is

$$P_{\pi} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{r}_{\pi} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \quad \gamma = \frac{1}{2}.$$

- a. Verify that P_{π} is stochastic and write all four scalar Bellman equations.
- b. Solve exactly for the four state values, working backward from the self-loop at s_4 .
- c. Starting from $\mathbf{v}^{(0)} = \mathbf{0}$, compute $\mathbf{v}^{(1)}$, $\mathbf{v}^{(2)}$, and $\mathbf{v}^{(3)}$.
- d. Compute the largest absolute error after three iterations and compare it with the bound from Exercise 2.19.
- e. An action c , not selected by the policy at s_1 , gives immediate reward 0, then moves to s_4 with probability 0.25 and to s_1 with probability 0.75. Compute $q_{\pi}(s_1, c)$.
- f. The policy selects one action with probability one at s_1 . That action gives reward 1, then moves to s_2 or s_3 with equal probability. Compute its action value and reconcile it with $v_{\pi}(s_1)$.

Chapter 3: Optimal State Values and the Bellman Optimality Equation

This chapter uses optimal state values, greedy policies, the Bellman optimality equation (BOE), contraction mappings, and the effects of rewards, discounting, and transition models. Algorithms whose implementation belongs to Chapter 4 are not assumed.

Part A: Optimality and Greedy Improvement

Exercise 3.1. Comparing policies state by state

(Numerical/derivation; Basic)

Three policies have state-value vectors, in the state order (s_1, s_2, s_3) ,

$$v_{\pi_1} = (4, 2, 5)^T, \quad v_{\pi_2} = (4, 3, 4)^T, \quad v_{\pi_3} = (5, 3, 5)^T.$$

For every ordered pair of policies, determine whether the first is better than the second under the chapter's componentwise definition. Identify every policy that could be optimal among this set, and explain why comparing only the sums of the vector entries would give a different, invalid criterion.

Exercise 3.2. Optimality is a partial order

(Conceptual/theory; Basic)

Explain why two policies need not be comparable under the definition $v_{\pi_1}(s) \geq v_{\pi_2}(s)$ for every state. Construct a two-state example of incomparable value vectors and discuss why an optimal policy, if present, must nevertheless dominate both vectors.

Exercise 3.3. A one-state greedy improvement

(Numerical/derivation; Basic)

At state s , the action values under a policy π are

$$q_{\pi}(s, a_1) = 1.8, \quad q_{\pi}(s, a_2) = 2.5, \quad q_{\pi}(s, a_3) = 2.1,$$

and $\pi(\cdot | s) = (0.5, 0.2, 0.3)$.

- Calculate $v_{\pi}(s)$.
- Write a deterministic greedy policy at s and calculate the weighted action-value expression it maximizes.
- Write every stochastic policy supported only on maximizing actions.
- Repeat part *c* if $q_{\pi}(s, a_3)$ is changed to 2.5.

Exercise 3.4. Why a probability mixture cannot beat its largest entry

(Numerical/derivation; Intermediate)

Let $q_1, \dots, q_m \in \mathbb{R}$ and let $p_i \geq 0$ with $\sum_{i=1}^m p_i = 1$.

- Prove that $\sum_i p_i q_i \leq \max_i q_i$.
- Characterize all probability vectors p for which equality holds.
- Apply the result to explain why the maximization over policies on the right-hand side of the BOE can be replaced by a maximization over actions.

Exercise 3.5. Unique value, nonunique policy*(Conceptual/theory; Intermediate)*

The BOE has a unique value solution, yet an MDP may have many optimal policies. Explain how these statements are compatible. Address deterministic ties, stochastic mixtures over tied actions, and why mixing in a strictly inferior action destroys optimality at the affected state.

Part B: Constructing and Solving Small Bellman Optimality Equations**Exercise 3.6. One state and three actions***(Numerical/derivation; Basic)*

A continuing MDP has one state s . Every action returns to s , and the deterministic rewards for a, b, c are 2, -1 , 1.5, respectively. Let $\gamma = 0.7$.

- Write the BOE for $v(s)$.
- Solve for $v_*(s)$ without assuming the optimal action in advance.
- Compute all three optimal action values $q_*(s, \cdot)$.
- List all optimal policies.

Exercise 3.7. A two-state choice with delayed reward*(Numerical/derivation; Intermediate)*

There are two states x, y . At y , the only action returns to y with reward 3. At x , action a returns to x with reward 1, while action b moves to y with reward -2 . Let $\gamma = 0.8$.

- Write the two BOE equations.
- Solve $v_*(y)$, then compare the two candidate action values at x .
- Solve $v_*(x)$ and identify the optimal action.
- Find the range of $\gamma \in (0, 1)$ for which action b is optimal at x .

Exercise 3.8. Stochastic transition in the BOE*(Numerical/derivation; Intermediate)*

At state s , action a gives reward 1 and moves to u with probability 0.75 and to v with probability 0.25. Action b gives reward 2 and moves to each state with probability 0.5. Suppose $v_*(u) = 8$, $v_*(v) = 2$, and $\gamma = 0.6$.

- Calculate $q_*(s, a)$ and $q_*(s, b)$.
- Determine $v_*(s)$ and an optimal policy at s .
- Holding all else fixed, solve for the transition probability $p = \mathbb{P}(u \mid s, a)$ at which the two actions tie.

Exercise 3.9. Recovering a policy from a proposed optimal value*(Numerical/derivation; Intermediate)*

For a two-state MDP with $\gamma = 0.5$, the reward vectors and transition matrices of two deterministic policies are

$$r_A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad P_A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad r_B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad P_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Given the proposed vector $v = (2, 3)^T$, calculate $r_A + \gamma P_A v$ and $r_B + \gamma P_B v$. Take the maximum componentwise, determine whether v is a fixed point of the BOE operator, and state which policy choice is greedy at each state.

Exercise 3.10. Verifying the special Bellman equation

(Numerical/derivation; Intermediate)

Suppose a vector v_* satisfies the BOE and a policy π_* attains the maximum in every component:

$$\pi_* \in \arg \max_{\pi} (r_{\pi} + \gamma P_{\pi} v_*).$$

Derive, step by step, that $v_* = r_{\pi_*} + \gamma P_{\pi_*} v_*$. Then use uniqueness of the solution of the ordinary Bellman equation for a fixed policy to show $v_* = v_{\pi_*}$. Clearly identify which conclusion still requires the optimality theorem rather than this algebra alone.

Exercise 3.11. Evaluating the BOE operator

(Numerical/derivation; Basic)

For

$$r_A = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad P_A = \begin{bmatrix} 1 & 0 \\ 0.4 & 0.6 \end{bmatrix}, \quad r_B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad P_B = \begin{bmatrix} 0 & 1 \\ 0.7 & 0.3 \end{bmatrix},$$

let $f(v) = \max\{r_A + 0.5P_A v, r_B + 0.5P_B v\}$, where the maximum is componentwise. Calculate $f(v)$ at $v = (0, 0)^T$, $(2, -1)^T$, and $(4, 4)^T$, and report the greedy policy associated with each component in each case.

Exercise 3.12. Bellman equation or Bellman optimality equation?

(Conceptual/theory; Basic)

For each expression below, state whether it is an ordinary Bellman equation, a BOE, or neither, and justify the classification:

$$v = r_{\pi} + \gamma P_{\pi} v, \quad v = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v), \quad v = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v_{\pi}), \quad q(s, a) = r(s, a) + \gamma v(s').$$

Discuss which quantities are fixed and which are unknown in each valid equation.

Part C: Fixed Points and Contraction Arguments

Exercise 3.13. Scalar fixed-point calculations

(Numerical/derivation; Basic)

For each map, find every fixed point and determine whether it is a contraction on $\mathbb{R}$ under absolute value:

$$f_1(x) = 0.4x + 3, \quad f_2(x) = 1.1x - 1, \quad f_3(x) = 0.6 \sin x.$$

For f_1 , start from $x_0 = -2$, compute x_1, x_2, x_3 , and compare the errors with the contraction bound.

Exercise 3.14. Iteration count from a contraction bound

(Numerical/derivation; Intermediate)

An iteration $x_{k+1} = f(x_k)$ has contraction factor $\gamma = 0.75$, and $|x_1 - x_0| = 12$.

- Use the Cauchy-sequence argument to derive $|x_* - x_n| \leq \gamma^n |x_1 - x_0| / (1 - \gamma)$.
- Find the smallest n for which this upper bound is below 0.01.

- c. Explain why the bound may be conservative even though it is valid.

Exercise 3.15. Empirical BOE error and a certified bound

(Numerical/derivation; Challenging)

The BOE operator has discount rate $\gamma = 0.9$. Two consecutive iterates satisfy $\|v_{k+1} - v_k\|_\infty = 0.04$.

- Derive an upper bound on $\|v_* - v_k\|_\infty$ using only this difference and the contraction property.
- Derive a bound on $\|v_* - v_{k+1}\|_\infty$.
- How would both bounds change if $\gamma = 0.5$? Interpret the effect of discounting on fixed-point convergence.

Exercise 3.16. Proving contraction of the maximum operator

(Numerical/derivation; Challenging)

Let

$$(Tv)(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s' | s, a) v(s') \right].$$

Prove from first principles that $\|Tu - Tv\|_\infty \leq \gamma \|u - v\|_\infty$. Your proof must explicitly establish the inequality $|\max_i x_i - \max_i y_i| \leq \max_i |x_i - y_i|$ and use the nonnegativity and unit row sum of transition probabilities.

Exercise 3.17. Why the maximum norm is natural

(Conceptual/theory; Challenging)

The standard BOE contraction proof uses the maximum norm. Explain precisely which step relies on it. Is every stochastic matrix guaranteed to have induced Euclidean norm at most one? Give either a proof or a two-state counterexample, and explain why failure of that claim does not invalidate the maximum-norm result.

Exercise 3.18. What the contraction theorem does and does not guarantee

(Conceptual/theory; Intermediate)

Separate the following claims into guaranteed, not guaranteed, or requiring an additional argument: existence of a BOE value solution; uniqueness of that value; uniqueness of the maximizing policy; convergence from every initial value; equality of every intermediate iterate to a policy value; and existence of a deterministic optimal policy. Justify each classification.

Part D: Optimal Policies and Sensitivity

Exercise 3.19. Proving greedy optimality after solving the BOE

(Numerical/derivation; Challenging)

Let v_* solve the BOE, and define

$$q_*(s, a) = \mathbb{E}[R_{t+1} | s, a] + \gamma \sum_{s'} p(s' | s, a) v_*(s').$$

Construct a deterministic policy by choosing one maximizer of $q_*(s, a)$ at each state. Prove that its Bellman equation is satisfied by v_* , and then prove that this policy is optimal. State where finiteness of the action sets is used.

Exercise 3.20. Building infinitely many optimal policies*(Numerical/derivation; Intermediate)*

Construct a two-state continuing MDP with two actions per state for which:

- the optimal state-value vector is unique;
- both actions tie at s_1 , but only one action is optimal at s_2 ;
- there are infinitely many stochastic optimal policies.

Specify all rewards and deterministic transitions, solve the BOE, and parameterize the full family of optimal policies.

Exercise 3.21. Why local greediness must use the correct value*(Conceptual/theory; Intermediate)*

A policy is chosen greedily with respect to an arbitrary vector u , not with respect to v_* or its own value. Explain why this fact alone does not prove that the policy is optimal. Describe the fixed-point or Bellman consistency condition that is missing and give a small qualitative counterexample.

Exercise 3.22. A discount-rate threshold between two routes*(Numerical/derivation; Intermediate)*

From s , a safe route gives rewards $(2, 2, 0, 0, \dots)$. A risky route gives rewards $(-1, 0, 8, 0, 0, \dots)$. The routes are deterministic and terminal after their listed nonzero rewards.

- Derive both returns as functions of γ .
- Find every $\gamma \in [0, 1)$ for which the risky route is optimal.
- Interpret the boundary values and describe how the result illustrates short-sighted versus far-sighted optimal behavior.

Exercise 3.23. How severe must a forbidden-state penalty be?*(Numerical/derivation; Challenging)*

Two deterministic choices start from the same state. Route A produces $(0, 0, 4, 4, 4, \dots)$. Route B produces $(-c, 5, 5, 5, \dots)$, where $c > 0$. Let $\gamma = 0.8$. Derive the exact threshold on c at which the preferred action changes. Then repeat symbolically for general $\gamma \in (0, 1)$ and discuss the limit as $\gamma \rightarrow 1$.

Exercise 3.24. Affine reward transformation*(Numerical/derivation; Basic)*

An MDP has $\gamma = 0.75$, optimal value $v_* = (4, -2, 6)^T$, and a known optimal policy. Every reward is transformed as $r' = 3r - 5$.

- Calculate the new optimal value vector without resolving the BOE.
- State what happens to the set of optimal policies.
- Verify your value transformation by substituting it into the transformed BOE in symbolic form.

Exercise 3.25. Proving reward-affine invariance*(Numerical/derivation; Challenging)*

Suppose every reward is transformed to $r' = \alpha r + \beta$, where $\alpha > 0$. Starting from the original BOE, derive

$$v'_* = \alpha v_* + \frac{\beta}{1 - \gamma} \mathbf{1}.$$

Then prove that the maximizing actions are unchanged. Your derivation must make explicit use of $P_\pi \mathbf{1} = \mathbf{1}$.

Exercise 3.26. What if the reward scale is negative?

(Conceptual/theory; Intermediate)

The affine-invariance theorem assumes $\alpha > 0$. Explain what fails if $\alpha < 0$. Construct a one-state, two-action counterexample and discuss whether $\alpha = 0$ preserves the original set of optimal policies.

Exercise 3.27. Discounting already penalizes a detour

(Numerical/derivation; Basic)

One policy reaches a state yielding reward 1 forever after one zero-reward transition. Another reaches the same state after four zero-reward transitions. For general γ , derive both returns, their difference, and their ratio. Evaluate the results at $\gamma = 0.9$, and explain why an extra per-step penalty is unnecessary for distinguishing these routes.

Exercise 3.28. The edge case $\gamma = 0$

(Conceptual/theory; Basic)

Describe exactly what the BOE becomes when $\gamma = 0$. Which aspects of the transition model cease to influence the optimal action? Can future-state information still matter indirectly through an immediate reward distribution? Give careful conditions for your answer.

Exercise 3.29. Sensitivity is not the same as affine invariance

(Conceptual/theory; Challenging)

Contrast three changes to an MDP: adding the same constant to every reward, changing only the penalty for one transition, and changing a transition probability. Explain why only the first is covered by the affine-invariance theorem and describe how each of the other two can reverse the optimal action.

Exercise 3.30. Capstone: a risky shortcut under model uncertainty

(Numerical/derivation; Challenging)

At state s , action A gives reward 1 and then enters a state worth 6. Action B gives reward -2 ; with probability p it enters a state worth 12, and with probability $1 - p$ a state worth -4 .

- Derive both optimal-action candidates as functions of p and γ .
- Solve for the boundary $p_*(\gamma)$ at which the actions tie.
- Determine the preferred action at $(p, \gamma) = (0.7, 0.8)$.
- Analyze how $p_*(\gamma)$ changes with γ , including its limiting behavior as $\gamma \downarrow 0$.
- Apply a positive affine reward transformation and verify directly that the boundary between the actions does not change.

Chapter 4: Value Iteration and Policy Iteration

This chapter covers value iteration, policy iteration, truncated policy iteration, their convergence logic, and generalized policy iteration. All problems in this chapter assume the environment model is available.

Part A: Value Iteration

Exercise 4.1. One complete value-iteration sweep

(Numerical/derivation; Basic)

There are two states s_1, s_2 , two actions a, b , and $\gamma = 0.5$. The one-step lookahead expressions are

	a	b
s_1	$1 + 0.5v(s_1)$	$0 + 0.5v(s_2)$
s_2	$2 + 0.5v(s_1)$	$-1 + 0.5v(s_2)$

Starting from $v_0 = (0, 0)^T$, compute every $q_0(s, a)$, the greedy policy π_1 , and v_1 . Repeat for the next sweep to obtain π_2, v_2 .

Exercise 4.2. Are value-iteration iterates state values?

(Conceptual/theory; Basic)

Explain why a vector v_k produced by value iteration need not be the state value of π_k , π_{k+1} , or any policy. State the Bellman consistency equation that would have to hold for it to be a genuine policy value, and explain why convergence changes the conclusion.

Exercise 4.3. Synchronous updates on a deterministic chain

(Numerical/derivation; Basic)

States s_1, s_2, s_3 form a chain. From s_1 , action R moves to s_2 with reward 0, while S stays with reward -1 . From s_2 , R moves to s_3 with reward 0, while S stays with reward -1 . At s_3 , the only action stays with reward 2. Let $\gamma = 0.8$ and $v_0 = 0$. Perform four synchronous value-iteration sweeps and record the greedy action at every state after each sweep. Explain how information about the target propagates backward.

Exercise 4.4. A stochastic value update

(Numerical/derivation; Intermediate)

At state x , action a gives reward 1 and moves to x or y with probabilities 0.3, 0.7. Action b gives reward 0 and moves to x or y with probabilities 0.8, 0.2. Let $\gamma = 0.9$ and $v_k(x) = 4, v_k(y) = 7$. Calculate both one-step lookahead values, $v_{k+1}(x)$, and the new greedy action. Find the value of $v_k(y)$ at which the decision switches when $v_k(x) = 4$.

Exercise 4.5. Matrix and elementwise forms agree

(Numerical/derivation; Intermediate)

For two deterministic policies A, B , let

$$r_A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad P_A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad r_B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad P_B = I,$$

with $\gamma = 0.6$ and $v_k = (3, 1)^T$. Use the matrix form to compute $v_{k+1} = \max_{\pi}(r_{\pi} + \gamma P_{\pi} v_k)$. Then reproduce the same result by evaluating actions separately at each state, noting that the componentwise maximizer may combine rows from different deterministic policies.

Exercise 4.6. Ties during value iteration*(Conceptual/theory; Intermediate)*

If several actions maximize $q_k(s, a)$, explain why choosing any one of them does not change the value update at that sweep. Could different tie choices lead to different intermediate policies? Could they lead to different optimal value vectors? Could they lead to different final optimal policies?

Exercise 4.7. Contraction-rate prediction*(Numerical/derivation; Intermediate)*

For value iteration with $\gamma = 0.85$, suppose $\|v_0 - v_*\|_\infty \leq 50$.

- Derive a bound on $\|v_k - v_*\|_\infty$.
- Find the smallest k that guarantees an error below 0.1.
- Repeat for $\gamma = 0.5$ and compare the iteration counts.

Exercise 4.8. A residual-based stopping guarantee*(Numerical/derivation; Challenging)*

Let T be the Bellman optimality operator. Prove that

$$\|v - v_*\|_\infty \leq \frac{\|Tv - v\|_\infty}{1 - \gamma}.$$

If $\gamma = 0.95$, how small must the Bellman residual be to certify value error at most 0.02? Explain the difference between this residual and the difference of two arbitrary stored vectors.

Exercise 4.9. Greedy policy quality from an approximate value*(Numerical/derivation; Challenging)*

Let u approximate v_* with $\|u - v_*\|_\infty \leq \varepsilon$, and let π_u be greedy with respect to u . Show that each one-step action score computed from u differs from its score computed from v_* by at most $\gamma\varepsilon$. Derive a sufficient gap condition under which π_u is guaranteed to choose the same unique action as an optimal greedy policy at a state.

Exercise 4.10. When should value iteration stop?*(Conceptual/theory; Intermediate)*

Compare stopping because the greedy policy has remained unchanged for one sweep, because $\|v_{k+1} - v_k\|$ is small, and because a certified Bellman residual is small. Which condition directly controls value accuracy? Give a case in which a policy stabilizes before the values do.

Part B: Policy Evaluation and Policy Improvement**Exercise 4.11. Exact evaluation of a two-state policy***(Numerical/derivation; Basic)*

Under policy π_0 , state x moves to y with reward 1, and y moves to x with reward 3. Let $\gamma = 0.5$.

- Solve the two Bellman equations exactly.
- Starting from zero, perform three iterative policy-evaluation sweeps.
- Compare the iterates with the exact values and identify the contraction factor.

Exercise 4.12. An iteration nested inside an iteration*(Conceptual/theory; Basic)*

Explain why standard policy iteration contains an inner iterative process and an outer iterative process. Identify what is fixed and what changes in each. Why is a finite inner stopping rule unavoidable in computation, and how does this observation motivate truncated policy iteration?

Exercise 4.13. A complete policy-improvement step*(Numerical/derivation; Basic)*

The evaluated value of the current policy is $v_{\pi_k}(s_1) = 2, v_{\pi_k}(s_2) = 5$. With $\gamma = 0.8$, the available one-step models are

	a	b
s_1	$(r = 1, s' = s_1)$	$(r = -1, s' = s_2)$
s_2	$(r = 0, s' = s_1)$	$(r = 2, s' = s_2)$.

Compute every $q_{\pi_k}(s, a)$, obtain π_{k+1} , and calculate the one-step backup $T_{\pi_{k+1}}v_{\pi_k}$. Verify that it is componentwise no smaller than v_{π_k} .

Exercise 4.14. Proving the policy-improvement lemma*(Numerical/derivation; Challenging)*

Starting from $\pi' = \arg \max_{\pi} (r_{\pi} + \gamma P_{\pi} v_{\pi})$, prove that $v_{\pi'} \geq v_{\pi}$. Give both of the following derivations:

- an iterative inequality using nonnegativity of $P_{\pi'}$ and $\gamma^n P_{\pi'}^n \rightarrow 0$;
- a derivation using $(I - \gamma P_{\pi'})^{-1} = \sum_{j=0}^{\infty} \gamma^j P_{\pi'}^j$.

Exercise 4.15. Two outer policy-iteration steps*(Numerical/derivation; Intermediate)*

Consider two states x, y with $\gamma = 0.5$. At x , action a stays at x with reward 0, while b moves to y with reward -1 . At y , action a moves to x with reward 4, while b stays at y with reward 1. Start from the policy choosing a at both states. Evaluate it exactly, improve it, evaluate the new policy, and improve once more. Determine whether the policy has become stable and verify its value against the BOE.

Exercise 4.16. Policy iteration in matrix form*(Numerical/derivation; Intermediate)*

For a three-state current policy, suppose

$$r_{\pi_k} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad P_{\pi_k} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \gamma = 0.5.$$

Solve v_{π_k} exactly. A candidate improved policy has

$$r_{\pi'} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \quad P_{\pi'} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Calculate $T_{\pi'}v_{\pi_k}$ and decide whether the policy-improvement inequality needed by the algorithm holds.

Exercise 4.17. Monotone values and policy convergence*(Conceptual/theory; Challenging)*

Policy iteration produces a nondecreasing sequence of value vectors bounded by v_* . Explain why this implies convergence of the values. Why does value convergence alone require care when claiming convergence of a particular policy sequence if ties are broken differently on different iterations?

Exercise 4.18. Policy iteration versus value iteration by sandwiching*(Numerical/derivation; Challenging)*

Assume $v_0 \leq v_{\pi_0}$. Reconstruct the induction showing

$$v_k \leq v_{\pi_k} \leq v_*,$$

where v_k is generated by value iteration and v_{π_k} by policy iteration. State exactly where policy improvement and nonnegativity of transition matrices enter the proof, and use the result to establish convergence of policy-iteration values.

Exercise 4.19. Work required by exact policy evaluation*(Numerical/derivation; Intermediate)*

A policy-evaluation operator has contraction factor $\gamma = 0.9$. Its initial error is at most 20 in maximum norm.

- How many inner sweeps guarantee error below 10^{-3} ?
- How many guarantee error below 10^{-1} ?
- If an outer algorithm performs 15 policy improvements, compare the resulting upper bounds on total inner sweeps for these tolerances.
- Explain why more accurate evaluation need not minimize total work.

Exercise 4.20. Why these algorithms are model based*(Conceptual/theory; Basic)*

Identify every place in value iteration and policy iteration where $p(r \mid s, a)$ or $p(s' \mid s, a)$ is used. Explain why observing one sampled transition is not the same as having these probability models, and state which later chapter begins replacing model-based expectations by sampled estimates.

Part C: Truncated Policy Iteration**Exercise 4.21. One, three, and infinitely many evaluation sweeps***(Numerical/derivation; Basic)*

For a fixed policy, the Bellman evaluation map is

$$T_\pi v = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0.5 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} v.$$

Starting from $v^{(0)} = (0, 0)^T$, compute $v^{(1)}$ and $v^{(3)}$, and solve for $v^{(\infty)} = v_\pi$. Interpret these as evaluation stages within value iteration, truncated policy iteration with truncation level 3, and policy iteration.

Exercise 4.22. A truncated policy-iteration outer step*(Numerical/derivation; Intermediate)*

Use the MDP from Exercise 4.15. Begin with the same initial policy but replace exact evaluation by two synchronous policy-evaluation sweeps initialized at zero. Improve greedily using this approximation. Compare the resulting policy with the one obtained after exact evaluation and explain any agreement or disagreement through action-value gaps.

Exercise 4.23. Approximate values are not policy values*(Conceptual/theory; Intermediate)*

In truncated policy iteration, why is the vector after finitely many evaluation sweeps generally not v_{π_k} ? Is it necessarily the value of some other policy? Explain why it can still be useful for policy improvement despite lacking exact Bellman consistency.

Exercise 4.24. Proving monotonic inner evaluation under a special initialization*(Numerical/derivation; Challenging)*

Let π_k be greedy with respect to $v_{\pi_{k-1}}$, and initialize the inner evaluation by $v_{\pi_k}^{(0)} = v_{\pi_{k-1}}$. Prove that $v_{\pi_k}^{(j+1)} \geq v_{\pi_k}^{(j)}$ for every $j \geq 0$. Begin by proving the first-step inequality and then propagate it using nonnegativity of P_{π_k} .

Exercise 4.25. Choosing a truncation level from an error target*(Numerical/derivation; Intermediate)*

For each outer policy, the initial policy-evaluation error is bounded by 8, and $\gamma = 0.75$. Find the smallest truncation level j that guarantees the remaining evaluation error is below 0.5, 0.05, and 0.005. Discuss how the marginal benefit of increasing j changes.

Exercise 4.26. Cost model for the three algorithms*(Numerical/derivation; Challenging)*

Suppose one full value backup costs one unit and one greedy policy update costs two units. Value iteration needs 80 outer sweeps. Truncated policy iteration with j evaluation sweeps needs N_j outer steps, where $N_1 = 80, N_2 = 50, N_4 = 32, N_8 = 25$. Exact policy iteration needs 18 outer steps and 30 evaluation sweeps per step.

- Calculate total cost for every listed method.
- Identify the cheapest choice under this model.
- Recalculate if a greedy policy update costs 15 units.
- Explain why no truncation level is universally best.

Exercise 4.27. Accuracy versus computation in policy evaluation*(Conceptual/theory; Intermediate)*

Discuss the competing effects of increasing the number of inner evaluation sweeps: accuracy of the current policy value, quality of the next greedy step, cost per outer iteration, and number of outer iterations. Explain why “evaluate as accurately as possible” is not a complete computational rule.

Exercise 4.28. Value iteration and policy iteration as endpoints*(Numerical/derivation; Intermediate)*

Write the truncated policy-iteration update with truncation level j . Algebraically specialize it to $j = 1$ and show the correspondence with value iteration. Then take $j \rightarrow \infty$, identify the fixed policy value reached by the inner loop, and show the correspondence with policy iteration.

Exercise 4.29. Capstone: compare the first decisions of three methods*(Numerical/derivation; Challenging)*

Consider a three-state MDP with two actions at each state and $\gamma = 0.8$:

	a	b
s_1	$(0, s_2)$	$(1, s_1)$
s_2	$(0, s_3)$	$(2, s_1)$
s_3	$(3, s_3)$	$(-1, s_2)$

Start value iteration from zero. Start policy iteration from the all- b policy. Start truncated policy iteration from the same policy, using two evaluation sweeps initialized at zero.

- Carry out the first complete outer step of each method.
- Compare their resulting value vectors and greedy policies.
- Continue each method until you can certify the optimal policy, showing all intermediate calculations.
- Explain the differences through the amount of policy evaluation each method performs.

Exercise 4.30. Generalized policy iteration as an organizing idea*(Conceptual/theory; Challenging)*

Explain generalized policy iteration without treating it as a single fixed algorithm. Describe how value iteration, policy iteration, and truncated policy iteration instantiate different schedules of evaluation and improvement. What can go wrong if either process is stopped permanently while the other continues?

Chapter 5: Monte Carlo Methods

This chapter introduces model-free Monte Carlo estimation, MC Basic, MC Exploring Starts, first-visit and every-visit sample use, MC ε -Greedy, and the exploration–exploitation tradeoff. The incremental stochastic-approximation machinery of Chapter 6 is not assumed.

Part A: Learning Expectations from Samples

Exercise 5.1. Model-based mean and sample-based mean

(Numerical/derivation; Basic)

A random return G takes values $-2, 1, 5$ with probabilities $0.2, 0.5, 0.3$.

- Calculate $\mathbb{E}[G]$ from the probability model.
- A sample is $(-2, 1, 1, 5, 1, 5, 5, 1)$. Calculate its Monte Carlo mean.
- Calculate the error of the sample mean and explain why a nonzero error does not make the estimator biased.
- If every sampled value is shifted upward by 4, determine how the true mean, sample mean, and estimation error change.

Exercise 5.2. Why the sampling assumptions matter

(Conceptual/theory; Basic)

The law-of-large-numbers argument in this chapter assumes independent, identically distributed samples. Explain the role of each word. Analyze the extreme sampling rule $X_i = X_1$ for all i : is the sequence identically distributed, is it independent, and does its average necessarily converge to the population mean?

Exercise 5.3. Mean and variance of a Monte Carlo average

(Numerical/derivation; Intermediate)

Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{var}(X_i) = 36$.

- Derive the expectation and variance of $\bar{X}_n = n^{-1} \sum_i X_i$.
- Find the smallest n for which the standard deviation of the mean is at most 0.3.
- By what factor must n increase to halve this standard deviation?
- Explain the difference between reducing estimator variance and changing its expected value.

Exercise 5.4. Combining two independent batches

(Numerical/derivation; Intermediate)

One independent batch has 40 returns with mean 3.2; another has 60 returns with mean 4.5.

- Derive the combined mean without reconstructing individual returns.
- Show why the unweighted average of the two batch means is generally incorrect.
- If every return has variance 25, calculate the variance of each batch mean and of the combined mean.

Exercise 5.5. Model, data, or neither

(Conceptual/theory; Basic)

Compare the information needed to calculate an expectation from a probability model with the information needed for a Monte Carlo estimate. What can be done if the model is unknown but

samples are available? What can be done if the model is known but no samples have been collected? Why is neither route available if both are absent?

Part B: MC Basic and Return Estimation

Exercise 5.6. Estimating one action value from episodes

(Numerical/derivation; Basic)

Starting from (s, a) and then following a fixed policy produces four three-step reward sequences:

$$(2, 0, 3), \quad (1, 2, 1), \quad (-1, 4, 2), \quad (3, -1, 0).$$

The episode terminates after the third reward and $\gamma = 0.8$. Calculate every return and the Monte Carlo estimate of $q_\pi(s, a)$. Then recalculate the estimate if the first episode is accidentally counted twice, and explain the source of the change.

Exercise 5.7. One MC Basic policy-improvement step

(Numerical/derivation; Basic)

For one state, the sampled returns are

$$\begin{array}{c|l} a_1 & 2, 5, 1, 4 \\ a_2 & 3, 3, 4, 2 \\ a_3 & -1, 7, 0, 1 \end{array}$$

under the current policy after the specified starting action.

- Estimate all action values.
- Obtain the greedy improved policy.
- Add one additional return of 12 for a_3 , recompute, and explain why limited data can change the improvement decision sharply.

Exercise 5.8. Why MC control estimates action values directly

(Conceptual/theory; Intermediate)

Explain why estimating only state values is insufficient for model-free greedy policy improvement when the transition and reward models are unknown. Why can action values be compared directly without those models? Contrast this with the model-based calculation used by policy iteration.

Exercise 5.9. Sampling error can reverse an action ranking

(Numerical/derivation; Intermediate)

Two actions have true return means $q_\pi(s, a) = 4$ and $q_\pi(s, b) = 4.5$. A small experiment observes returns

$$a : 3, 7, 5, 4, 6, \quad b : 2, 3, 8, 2, 5.$$

Calculate the estimates and the resulting greedy choice. Construct one additional return for each action that reverses that sample-based choice while leaving the true ordering unchanged. Explain why the law of large numbers is an asymptotic, not finite-sample, guarantee.

Exercise 5.10. Truncating a continuing return

(Numerical/derivation; Intermediate)

An episode prefix produces reward 0 for L steps and reward 2 forever afterward. The true continuing return starts before the zero rewards.

- Derive the true return as a function of L and γ .
- What return is recorded if the generated episode is truncated before the first positive reward?
- At $\gamma = 0.9$, calculate the true return for $L = 5, 15, 30$.
- Explain how insufficient episode length can make a useful action appear worthless.

Exercise 5.11. Sparse rewards and episode length

(Conceptual/theory; Intermediate)

In a grid world with rewards only near a distant target, explain why short episodes first give nonzero estimates to states close to the target. What spatial pattern should emerge as episode length increases? Distinguish this truncation problem from randomness caused by too few independent episodes.

Part C: First-Visit, Every-Visit, and Exploring Starts

Exercise 5.12. Returns from repeated visits in one episode

(Numerical/derivation; Basic)

An episode is

$$s_0, a_0, 1, \quad s_1, a_1, 2, \quad s_0, a_0, -1, \quad s_2, a_2, 4,$$

and terminates after the last reward. Let $\gamma = 0.5$.

- Work backward to calculate the return following every time step.
- Give the first-visit sample for (s_0, a_0) .
- Give all every-visit samples for (s_0, a_0) and their within-episode average.
- Identify the initial-visit sample used if only the episode's starting pair is credited.

Exercise 5.13. Updating return totals and visit counts

(Numerical/derivation; Basic)

Before an episode, a table stores $\text{Returns}(s, a) = 18$ and $\text{Num}(s, a) = 6$. In the new episode, every-visit processing yields returns 3.5, 5.0, 1.5 for the same pair.

- Update both stored quantities after each visit.
- Record the action-value estimate after each update.
- Perform the corresponding first-visit update and compare the results.

Exercise 5.14. First-visit versus every-visit reasoning

(Conceptual/theory; Intermediate)

Compare first-visit and every-visit Monte Carlo estimation in terms of how much of an episode they use and how samples within one episode can be correlated. Why is every-visit often more sample-efficient in practice even though counting more returns from the same episode does not create more independent episodes?

Exercise 5.15. A deterministic exploring-starts schedule

(Numerical/derivation; Intermediate)

An MDP has 6 states, each with 4 actions. An experiment cycles through all starting state-action pairs equally often and generates 1200 episodes.

- a. How many episodes start from each pair?
- b. If 15% of the episodes starting from a particular pair terminate early and are discarded, how many usable samples remain for it?
- c. If return variance for that pair is 64, compare the standard deviation of its mean before and after discarding.

Exercise 5.16. Random exploring starts and coverage

(Numerical/derivation; Challenging)

There are M state–action pairs, and each episode starts from a uniformly selected pair, independently of other episodes.

- a. Derive the probability that a fixed pair is never selected in N episodes.
- b. Derive the expected number of pairs never selected.
- c. Evaluate both quantities for $M = 20, N = 100$.
- d. Explain why positive starting probability alone does not guarantee good finite-sample coverage.

Exercise 5.17. Initial-visit sample efficiency

(Numerical/derivation; Intermediate)

Ten episodes each contain 50 time steps. Under initial-visit processing, exactly one return per episode is used. Under first-visit processing, an average of 12 distinct state–action pairs occur per episode. Under every-visit processing, all 50 visits are used. Calculate the number of credited returns for each strategy. Then explain why these raw counts do not by themselves determine estimator variance or policy quality.

Exercise 5.18. What exploring starts assumes

(Conceptual/theory; Intermediate)

State the exploring-starts condition in operational terms. Why is it strong or even infeasible in some physical systems? Explain how failure to start from a pair differs from visiting that pair later in a sufficiently exploratory episode, and identify the idea used by MC ε -Greedy to remove the condition.

Part D: ε -Greedy Policies

Exercise 5.19. Action probabilities under ε -greedy selection

(Numerical/derivation; Basic)

A state has five actions and a unique greedy action.

- a. Derive the probability of the greedy action and of each nongreedy action under the chapter's ε -greedy rule.
- b. Evaluate them at $\varepsilon = 0, 0.2, 0.7, 1$.
- c. Verify that the probabilities sum to one and that the greedy action is never less probable than a nongreedy action.

Exercise 5.20. Sampling an ε -greedy action

(Numerical/derivation; Basic)

There are four actions, a_2 is greedy, and $\varepsilon = 0.3$. Actions are ordered a_1, a_2, a_3, a_4 . Construct explicit subintervals of $[0, 1)$ whose lengths equal the action probabilities, so a uniform random num-

ber selects an action. Use your mapping to choose actions for random numbers 0.04, 0.28, 0.66, 0.94. Then construct the alternative two-stage “exploit or sample uniformly” implementation and verify the same marginal probabilities.

Exercise 5.21. Optimization over the ε -greedy policy class

(Numerical/derivation; Intermediate)

At a four-action state, estimated action values are $(6, 2, 5, -1)$ and $\varepsilon = 0.4$.

- Write the optimal ε -greedy distribution.
- Calculate its expected action value.
- Compare it with the greedy value and the uniform-policy value.
- Repeat if the first and third actions tie at value 6, and describe all maximizing ε -greedy choices under a fixed tie rule.

Exercise 5.22. Optimal in Π_ε is not optimal in Π

(Conceptual/theory; Intermediate)

Explain the two meanings of saying that MC ε -Greedy “can” and “cannot” find an optimal policy. Distinguish optimality within the restricted class Π_ε from unrestricted optimality, and explain what happens as ε becomes small but remains positive.

Exercise 5.23. Expected visit counts under a soft policy

(Numerical/derivation; Intermediate)

At a recurrent state with four actions, suppose the state is visited 2000 times and the same action remains uniquely greedy. For $\varepsilon = 0.1, 0.5, 1$, calculate the expected count for the greedy action and each other action. Discuss which setting supplies the smallest action with the most data and which most heavily exploits the current estimate.

Exercise 5.24. Value loss caused by persistent exploration

(Numerical/derivation; Intermediate)

In a one-state continuing MDP, all actions return to the state. The immediate rewards for three actions are 3, 1, -2 , and $\gamma = 0.8$.

- Derive the value of the greedy policy.
- Derive the value of the ε -greedy policy as a function of ε .
- Evaluate its loss relative to greedy at $\varepsilon = 0.1, 0.4, 1$.
- Explain why exploration can reduce the final policy’s value even while improving learning coverage.

Exercise 5.25. An ε threshold near a dangerous state

(Numerical/derivation; Challenging)

At state s , staying yields immediate reward 5, while each of three other actions yields -20 ; every action then returns to s . Derive the value of the ε -greedy policy and find the largest ε for which its value is nonnegative when $\gamma = 0.9$. Interpret the result as a cost of random exploration near a hazardous region.

Part E: Exploration, Exploitation, and Synthesis

Exercise 5.26. The exploration–exploitation tension

(Conceptual/theory; Basic)

Explain why a policy that always selects the currently estimated best action may never discover a better one, while a uniformly random policy may learn values broadly but perform poorly. Describe how ε -greedy selection interpolates between these extremes and why no fixed ε is ideal for every stage of learning.

Exercise 5.27. Comparing two exploration schedules

(Numerical/derivation; Intermediate)

For 1000 visits to a state with five actions and a stable unique greedy action, compare these schedules:

$$\text{A: } \varepsilon_t = 0.5 \text{ for all } t, \quad \text{B: } \varepsilon_t = \begin{cases} 1, & t \leq 200, \\ 0.1, & t > 200. \end{cases}$$

Calculate the expected total number of selections of the greedy action and of each nongreedy action under both schedules. State which schedule explores more overall and which exploits more overall.

Exercise 5.28. A complete every-visit MC ε -greedy update

(Numerical/derivation; Challenging)

An episode with $\gamma = 0.5$ is

$$s, a, 2, \quad u, b, -1, \quad s, c, 4, \quad u, b, 3,$$

and then terminates. Initially all return totals and counts are zero.

- Compute the return after every visit by working backward.
- Update the every-visit totals, counts, and action-value estimates for every observed pair.
- At u , suppose a third action d has existing estimate 1.5. With $\varepsilon = 0.3$, write the improved policy at u .
- Explain why the policy used to generate the next episode must retain nonzero probabilities for nongreedy actions.

Exercise 5.29. How many episodes reduce Monte Carlo variability?

(Numerical/derivation; Intermediate)

Returns from a fixed state–action pair have variance 100. Assuming i.i.d. episodes, calculate the standard deviation of the Monte Carlo mean after $n = 4, 25, 100, 400$ episodes. How many episodes are needed for standard deviation at most 0.2? If returns from repeated visits within an episode are positively correlated, explain why using the same formula with the raw visit count may be optimistic.

Exercise 5.30. What if exploration is removed too early?

(Conceptual/theory; Challenging)

An MC control procedure begins with a large ε but sets it to zero after ten episodes. Analyze the possible consequences if an initially underestimated action has not yet been sampled. Conversely, what is lost if ε never decreases? Give a qualitative set of conditions a decaying exploration scheme should balance, without invoking algorithms from later chapters.

Chapter 6: Stochastic Approximation

This chapter covers incremental mean estimation, Robbins–Monro root finding, step-size conditions, Dvoretzky-type convergence arguments, stochastic gradient descent, and comparisons among batch, mini-batch, and stochastic gradient descent.

Part A: From Batch Averages to Incremental Updates

Exercise 6.1. Deriving the incremental sample mean

(Numerical/derivation; Basic)

Let $w_{k+1} = k^{-1} \sum_{i=1}^k x_i$ and $w_k = (k-1)^{-1} \sum_{i=1}^{k-1} x_i$. Derive

$$w_{k+1} = w_k + \frac{1}{k}(x_k - w_k).$$

Starting from $w_1 = x_1$, apply the recursion to $(x_1, x_2, x_3, x_4) = (6, 2, -1, 5)$ and verify after every step that it equals the corresponding ordinary sample average.

Exercise 6.2. What incrementality buys

(Conceptual/theory; Basic)

Compare batch and incremental mean calculation in terms of storage, waiting time, and availability of an interim estimate. Why can an early incremental estimate be useful even when it is inaccurate? Explain which information the recursion must retain from the past.

Exercise 6.3. Unrolling a constant-step mean estimator

(Numerical/derivation; Intermediate)

For $0 < \alpha < 1$, consider $w_{k+1} = (1 - \alpha)w_k + \alpha x_k$.

- Unroll the recursion to express w_{k+1} in terms of w_1 and $x_1, \dots, x_k$.
- Verify that the weights form a convex combination.
- For $\alpha = 0.2$, determine the weights on x_k, x_{k-1}, x_{k-5} .
- Explain how constant step size creates recency weighting rather than an ordinary sample mean.

Exercise 6.4. A general decaying step-size calculation

(Numerical/derivation; Basic)

Use $w_{k+1} = w_k - \alpha_k(w_k - x_k)$, $w_1 = 4$, observations $(x_1, x_2, x_3, x_4) = (1, 7, 3, 9)$, and $\alpha_k = 2/(k+2)$. Calculate $w_2, \dots, w_5$. Compare w_5 with the ordinary average of the four observations and explain why they need not be equal for this step-size sequence.

Exercise 6.5. Which power-law step sizes qualify?

(Numerical/derivation; Intermediate)

Let $\alpha_k = k^{-p}$ with $p > 0$. Determine all p for which

$$\sum_{k=1}^{\infty} \alpha_k = \infty, \quad \sum_{k=1}^{\infty} \alpha_k^2 < \infty.$$

Justify your answer using convergence of p -series, and classify $p = 0.4, 0.5, 0.75, 1, 1.2$.

Part B: Robbins–Monro Root Finding

Exercise 6.6. Noiseless Robbins–Monro steps

(Numerical/derivation; Basic)

Let $g(w) = 2w - 6$, whose root is $w_* = 3$. Use $w_{k+1} = w_k - a_k g(w_k)$, $w_1 = -1$, and $a_k = 1/(2k)$. Calculate the first five estimates, derive an error recursion for $e_k = w_k - w_*$, and obtain a closed expression for e_{k+1} .

Exercise 6.7. Why Robbins–Monro is a black-box method

(Conceptual/theory; Basic)

Explain what information Robbins–Monro needs about an unknown function $g(w)$, what it does not need, and how a noisy observation $\tilde{g}(w, \eta) = g(w) + \eta$ is used. Contrast it with a method that requires an analytical derivative of g .

Exercise 6.8. A noisy root-finding trace

(Numerical/derivation; Intermediate)

For $g(w) = w - 2$, use $w_1 = 5$, $a_k = 1/k$, and observed noises $(\eta_1, \eta_2, \eta_3, \eta_4) = (1, -2, 0.5, -1)$. Compute $w_2, \dots, w_5$ from

$$w_{k+1} = w_k - a_k [g(w_k) + \eta_k].$$

At each step separate the drift toward the root from the perturbation caused by noise.

Exercise 6.9. Monotonicity gives the correct direction

(Numerical/derivation; Intermediate)

Assume $g(w_*) = 0$ and g is strictly increasing.

- Show that $g(w) > 0$ when $w > w_*$ and $g(w) < 0$ when $w < w_*$.
- In the noiseless update $w^+ = w - ag(w)$, derive sufficient inequalities on $a|g(w)|$ that prevent overshooting the root.
- Apply them to $g(w) = \tanh(w - 1)$, $w = 3$, and $a = 0.4$.

Exercise 6.10. Interpreting the two step-size sums

(Conceptual/theory; Intermediate)

Explain the distinct roles of $\sum_k a_k = \infty$ and $\sum_k a_k^2 < \infty$. Relate the first to the ability to travel arbitrarily far from a poor initial guess and the second to suppressing persistent noise. Why does neither condition alone suffice?

Exercise 6.11. Testing Robbins–Monro assumptions

(Numerical/derivation; Intermediate)

For each function below, determine whether constants $0 < c_1 \leq c_2 < \infty$ exist such that $c_1 \leq g'(w) \leq c_2$ for all $w \in \mathbb{R}$:

$$g_1(w) = 3w + 1, \quad g_2(w) = w^3 - 2, \quad g_3(w) = \tanh w, \quad g_4(w) = w + 0.2 \sin w.$$

For every failure, identify whether the lower bound, upper bound, or both fail, and discuss what the theorem's failure does and does not imply about actual convergence from a particular initialization.

Exercise 6.12. Conditional zero-mean noise

(Numerical/derivation; Challenging)

Let the history H_k determine w_k , and suppose $\eta_k = Z_k \sigma(w_k)$, where Z_k is independent of H_k , $\mathbb{P}(Z_k = 1) = \mathbb{P}(Z_k = -1) = 1/2$, and $|\sigma(w)| \leq 5$. Compute $\mathbb{E}[\eta_k | H_k]$ and $\mathbb{E}[\eta_k^2 | H_k]$, and verify

the relevant noise assumptions. Then repeat if $\mathbb{P}(Z_k = 1 \mid H_k) = 0.8$ and explain which condition fails.

Exercise 6.13. Constant step size: useful but outside the theorem

(Conceptual/theory; Intermediate)

Why does a constant positive $a_k = a$ violate the square-summability condition? Describe the expected long-run behavior under persistent zero-mean noise and explain why constant step sizes are nevertheless used when the target may change over time.

Part C: Dvoretzky-Type Convergence Reasoning

Exercise 6.14. Deterministic contraction as a special process

(Numerical/derivation; Basic)

Consider $\Delta_{k+1} = (1 - \alpha_k)\Delta_k + \beta_k\eta_k$ with $\beta_k = 0$, $\alpha_k = 1/(k+1)$, and $\Delta_1 = 8$. Derive a closed expression for Δ_k , verify convergence to zero, and check the step-size sum conditions directly.

Exercise 6.15. Mean estimation in error coordinates

(Numerical/derivation; Intermediate)

Starting from $w_{k+1} = w_k + \alpha_k(X_k - w_k)$, let $w_* = \mathbb{E}[X]$ and $\Delta_k = w_k - w_*$.

- Derive the Dvoretzky form $\Delta_{k+1} = (1 - \alpha_k)\Delta_k + \alpha_k\eta_k$.
- Identify η_k .
- Verify its conditional mean and variance when X_k is i.i.d. with finite variance.
- State the conclusion when the step-size assumptions hold.

Exercise 6.16. Almost sure convergence and finite runs

(Conceptual/theory; Challenging)

Explain what an almost sure convergence statement says about an infinite random sequence and what it does not promise about any particular finite run. Why are conditional expectations taken with respect to the history? Discuss why rare sample paths do not contradict an almost sure theorem.

Exercise 6.17. A squared-error drift calculation

(Numerical/derivation; Challenging)

For $\Delta_{k+1} = (1 - \alpha_k)\Delta_k + \beta_k\eta_k$, define $h_k = \Delta_k^2$. Expand $h_{k+1} - h_k$, then calculate $\mathbb{E}[h_{k+1} - h_k \mid H_k]$ assuming $\mathbb{E}[\eta_k \mid H_k] = 0$ and $\mathbb{E}[\eta_k^2 \mid H_k] \leq C$. Identify the negative drift term and the noise term, and explain how square summability controls the latter.

Exercise 6.18. Mapping Robbins–Monro error into Dvoretzky form

(Numerical/derivation; Challenging)

Let w_* solve $g(w_*) = 0$, and define $\Delta_k = w_k - w_*$ for the Robbins–Monro update. Using the mean value theorem, derive

$$\Delta_{k+1} = [1 - a_k g'(\xi_k)]\Delta_k + a_k(-\eta_k)$$

for some ξ_k between w_k and w_* . Show how bounds on g' and the sums involving a_k produce the coefficient conditions required by the convergence theorem.

Part D: Stochastic Gradient Descent

Exercise 6.19. From expected loss to stochastic gradient

(Numerical/derivation; Basic)

Let $J(w) = \mathbb{E}[\frac{1}{2}(w - X)^2]$. Derive the true gradient and its root, write the gradient-descent update, and then replace the expectation by one sample to obtain SGD. For $w_1 = 0$, $\alpha_k = 1/k$, and samples 4, 2, 8, 6, compute the iterates and verify their connection to incremental mean estimation.

Exercise 6.20. Why a stochastic gradient can work

(Conceptual/theory; Intermediate)

Write a stochastic gradient as the true gradient plus a noise term. Explain why the noise has zero mean under i.i.d. sampling and why this property makes the update direction correct on average. Why does zero mean not imply that each individual update decreases the objective?

Exercise 6.21. SGD on a random quadratic

(Numerical/derivation; Intermediate)

Let $f(w, X) = \frac{1}{2}(cw - X)^2$, where $c \neq 0$ is known.

- Derive $\nabla_w f(w, X)$ and the SGD update.
- Solve analytically for the minimizer of $J(w) = \mathbb{E}[f(w, X)]$.
- With $c = 2$, samples 2, 6, -2, $w_1 = 1$, and constant $\alpha = 0.1$, compute three updates.
- Express the gradient noise and verify its mean is zero.

Exercise 6.22. Checking convexity and curvature bounds

(Numerical/derivation; Intermediate)

For each sample loss, calculate its second derivative with respect to w and determine whether fixed positive constants c_1, c_2 can bound it above and below for all $w, x \in \mathbb{R}$:

$$\frac{1}{2}(w - x)^2, \quad \frac{1}{4}(w - x)^4, \quad \frac{1}{2}e^x w^2 \text{ with } x \in [-1, 1], \quad \log(1 + e^{wx}).$$

State which losses meet the scalar curvature condition used by the chapter's SGD convergence theorem.

Exercise 6.23. Relative stochastic-gradient error

(Numerical/derivation; Challenging)

For mean estimation, the stochastic gradient is $(w - X)$, the true gradient is $w - \mu$, and the optimum is $w_* = \mu$.

- Derive the relative error $\delta = |X - \mu| / |w - \mu|$.
- Let $\mu = 2$ and $X = -1$. Evaluate δ at $w = 20, 5, 2.5, 2.1$.
- Explain quantitatively why SGD looks more like gradient descent far from the optimum and more random near it.

Exercise 6.24. Turning a finite dataset into a random variable

(Conceptual/theory; Intermediate)

For the deterministic objective $J(w) = n^{-1} \sum_{i=1}^n f(w, x_i)$, explain how introducing a uniformly distributed random variable $X \in \{x_1, \dots, x_n\}$ converts it to $\mathbb{E}[f(w, X)]$. Why must sampled indices be uniform and independent for the basic SGD argument? What changes if examples are sampled with unequal probabilities but the update is left uncorrected?

Exercise 6.25. Batch, mini-batch, and stochastic gradients*(Numerical/derivation; Basic)*

At a fixed w , four per-sample gradients are $-4, 2, 6, 0$.

- Calculate the full batch gradient.
- List every possible mini-batch gradient for batches of size two drawn without replacement.
- Calculate the mean and variance of those mini-batch gradients.
- Calculate the mean and variance of a one-sample stochastic gradient.
- Explain the variance reduction obtained by averaging.

Exercise 6.26. Sampling with and without replacement*(Numerical/derivation; Intermediate)*

A dataset contains values $0, 2, 8$, and a mini-batch of size three is drawn.

- If sampling is without replacement, find the mini-batch mean.
- If sampling is with replacement, list the possible means and their probabilities.
- Calculate the expected with-replacement mean and its variance.
- Use the result to explain why mini-batch size $m = n$ need not equal batch gradient descent when sampling permits repeats.

Exercise 6.27. Computational tradeoffs among BGD, MBGD, and SGD*(Conceptual/theory; Basic)*

Compare the three methods in terms of work per update, randomness of an update, memory access, and typical progress per update. Why can a method with noisier steps still reach a useful neighborhood sooner in wall-clock time? Why is “fewer iterations” not the same as “less computation”?

Exercise 6.28. A mini-batch mean recursion*(Numerical/derivation; Intermediate)*

At iteration k , a mini-batch mean is $\bar{x}_k^{(m)}$, and

$$w_{k+1} = w_k + \frac{1}{k}(\bar{x}_k^{(m)} - w_k).$$

Prove by induction that $w_{k+1} = k^{-1} \sum_{j=1}^k \bar{x}_j^{(m)}$. If the successive mini-batch means are $3, 5, 4, 8$, compute every iterate. Assuming independent sample variance σ^2 and batch size m , derive the variance of w_{k+1} .

Exercise 6.29. SGD as Robbins–Monro*(Numerical/derivation; Challenging)*

For $J(w) = \mathbb{E}[f(w, X)]$, define $g(w) = \nabla_w J(w)$ and $\tilde{g}(w, x) = \nabla_w f(w, x)$.

- Express $\tilde{g} = g + \eta$ and derive SGD as a Robbins–Monro update.
- Verify the conditional zero-mean property under i.i.d. sampling.
- Show how bounds on $\nabla_w^2 f$ imply bounds on $g'(w)$.
- Map every assumption of the chapter’s SGD convergence theorem to the corresponding Robbins–Monro assumption.

Exercise 6.30. Reading an SGD convergence trace*(Conceptual/theory; Challenging)*

An SGD trace moves quickly from a distant initial point toward the optimum, then fluctuates in a narrow region. Explain this pattern using the relative gradient error, sample variance, and step size. Predict qualitatively what happens when the mini-batch size increases, the step size decreases, or the sample distribution becomes more variable. Distinguish convergence of the theoretical decaying-step process from steady fluctuation under a constant step.

Chapter 7: Temporal-Difference Methods

This chapter covers tabular TD state-value learning, Sarsa, Expected Sarsa, n -step Sarsa, Q-learning, on-policy and off-policy learning, and the unified TD-target viewpoint. Function approximation is reserved for Chapter 8.

Part A: TD Learning of State Values

Exercise 7.1. A sequence of tabular TD updates

(Numerical/derivation; Basic)

Let $\gamma = 0.8$, $\alpha = 0.25$, and initial values $v_0(x) = 1, v_0(y) = 3$. A policy generates the transitions

$$(x, 2, y), \quad (y, -1, x), \quad (x, 0, y),$$

where each tuple is (s_t, r_{t+1}, s_{t+1}) . Apply the TD state-value update in chronological order. At every step report the TD target, TD error, updated value, and unchanged value.

Exercise 7.2. TD versus Monte Carlo

(Conceptual/theory; Basic)

Compare TD and Monte Carlo value learning in terms of when an update can be made, whether a terminal episode is required, bootstrapping, need for initial values, and the number of random future quantities in a target. Explain why the variance comparison does not imply that one method is uniformly better.

Exercise 7.3. The new estimate moves toward the TD target

(Numerical/derivation; Basic)

Write the update as $v^+ = v - \alpha(v - \bar{v})$, where $\bar{v} = r + \gamma v(s')$. Prove that

$$v^+ - \bar{v} = (1 - \alpha)(v - \bar{v}).$$

For $0 < \alpha < 1$, show that the absolute distance to the target strictly decreases unless the TD error is zero. Analyze what happens at $\alpha = 1$, $\alpha > 1$, and $\alpha < 0$.

Exercise 7.4. Expected TD error at the true value

(Numerical/derivation; Intermediate)

Starting from the Bellman expectation equation

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s],$$

derive

$$\mathbb{E}[v_\pi(S_t) - (R_{t+1} + \gamma v_\pi(S_{t+1})) \mid S_t = s] = 0.$$

Then give a two-outcome transition example in which the realized TD error is nonzero on every sample even though its conditional expectation is zero.

Exercise 7.5. What bootstrapping means

(Conceptual/theory; Intermediate)

Explain precisely why the TD target bootstraps and why a Monte Carlo return does not. How can an inaccurate value at s_{t+1} contaminate an update at s_t ? Conversely, how can repeated TD updates propagate useful information through a state sequence before any complete return is available?

Exercise 7.6. State visits and step-size conditions*(Numerical/derivation; Challenging)*

A state s is visited at global times $t_k = 2^k$, and on its k th visit the nonzero learning rate is $1/k$; at all other times $\alpha_t(s) = 0$.

- Determine whether $\sum_t \alpha_t(s)$ diverges and $\sum_t \alpha_t(s)^2$ converges.
- Repeat if the visit-time learning rate is $1/k^{0.4}$.
- Repeat if the state is visited only 100 times.
- Explain why the theorem's condition is state specific rather than a condition only on the global clock.

Part B: Sarsa and Expected Sarsa**Exercise 7.7. One Sarsa update***(Numerical/derivation; Basic)*

At time t ,

$$q_t(s_t, a_t) = 2.0, \quad r_{t+1} = -1, \quad q_t(s_{t+1}, a_{t+1}) = 5.0,$$

with $\gamma = 0.9$ and $\alpha = 0.2$. Calculate the Sarsa target, TD error under the book's sign convention $q - \text{target}$, and updated action value. Repeat if the sampled next action has value -2 , and explain why the behavior policy affects the update.

Exercise 7.8. Deriving the action-value Bellman expectation equation*(Numerical/derivation; Intermediate)*

Begin with

$$q_\pi(s, a) = \sum_r p(r \mid s, a) r + \gamma \sum_{s'} p(s' \mid s, a) \sum_{a'} \pi(a' \mid s') q_\pi(s', a').$$

Use conditional independence to rewrite it as $q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid s, a]$. Identify the single sampled quantity that replaces each expectation in Sarsa.

Exercise 7.9. Sarsa control as generalized policy iteration*(Conceptual/theory; Intermediate)*

In Sarsa control, the current policy is not evaluated to completion before it is improved. Explain how value estimation and policy improvement interact one sample at a time. Why is the policy made ϵ -greedy, and why is the new policy immediately used to generate subsequent actions?

Exercise 7.10. Expected Sarsa target*(Numerical/derivation; Basic)*

At s' , the current action values are $(4, 1, -2, 3)$, and the policy probabilities are $(0.7, 0.1, 0.1, 0.1)$. The observed reward is 2 and $\gamma = 0.75$.

- Calculate the Expected Sarsa target.
- List the possible ordinary Sarsa targets and their probabilities.
- Verify that the expected ordinary Sarsa target equals the Expected Sarsa target.
- Compute the variance of the ordinary Sarsa target arising only from the next-action choice.

Exercise 7.11. Sarsa with an ε -greedy next action*(Numerical/derivation; Intermediate)*

At s' , there are three actions with values 6, 3, 0, and the behavior policy is ε -greedy with $\varepsilon = 0.3$. Given $r = 1$ and $\gamma = 0.8$, calculate the probability distribution, mean, and variance of the Sarsa target. Compare its mean with the Q-learning target for the same transition.

Exercise 7.12. Decaying or constant learning rates during control*(Conceptual/theory; Challenging)*

The convergence theorem for evaluating a fixed policy uses decaying learning rates, while control implementations often use a small constant. Explain why the policy's continual change makes old estimates stale and why a vanishing rate can adapt too slowly. State the price paid for using a constant rate in a stationary late stage.

Part C: n -Step Sarsa**Exercise 7.13. Computing several n -step targets***(Numerical/derivation; Basic)*

Starting at time t , the rewards are $r_{t+1} = 1, r_{t+2} = 2, r_{t+3} = -1, r_{t+4} = 4, \gamma = 0.5$, and

$$q(s_{t+1}, a_{t+1}) = 6, \quad q(s_{t+2}, a_{t+2}) = 5, \quad q(s_{t+3}, a_{t+3}) = 2.$$

Calculate the one-, two-, and three-step Sarsa targets and the four-reward Monte Carlo return if the episode terminates after r_{t+4} .

Exercise 7.14. Why an n -step update is delayed*(Numerical/derivation; Intermediate)*

List the exact samples needed to form an n -step target for (s_t, a_t) . Show why they are unavailable at time t , and rewrite the update with its actual execution time $t + n$. For an episode of length T and $n = 4$, state at which times updates for $t = 0, 1, T - 4$ can first occur.

Exercise 7.15. Sarsa and Monte Carlo as endpoint cases*(Numerical/derivation; Intermediate)*

Starting from

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n q_\pi(S_{t+n}, A_{t+n}),$$

show algebraically how $n = 1$ gives the Sarsa target and how, for a terminating episode with n extending to the end, the bootstrap term disappears and the target becomes the Monte Carlo return. Explain the additional role of setting $\alpha = 1$ in the chapter's endpoint comparison.

Exercise 7.16. Bias and variance as n changes*(Conceptual/theory; Intermediate)*

Explain why a short n -step target relies more heavily on estimated values, while a long target relies on more sampled rewards. Relate these facts to bias and variance. Why can neither $n = 1$ nor a full Monte Carlo return be declared universally best?

Exercise 7.17. Choosing n in a deterministic example*(Numerical/derivation; Challenging)*

A deterministic episode has rewards 0, 0, 5, 0 and then terminates, with $\gamma = 0.9$. Current bootstrap estimates after the first, second, and third transitions are 3, 4, 1, respectively.

- Calculate the one-, two-, three-, and four-step targets at the initial pair.
- Calculate each target's error relative to the full return.
- If the estimates are all shifted upward by 2, repeat the errors.
- Explain how bootstrap quality, not only n , determines target bias.

Part D: Q-Learning and Policy Relationships

Exercise 7.18. A Q-learning table update

(Numerical/derivation; Basic)

For a transition (s, a, r, s') , let $q_t(s, a) = 1.5$, $r = -2$, $\gamma = 0.9$, $\alpha = 0.4$, and action values at s' be 4, 1, 3. Calculate the Q-learning target, signed TD error, and updated entry. State exactly which table entries remain unchanged.

Exercise 7.19. Behavior policy and target policy

(Conceptual/theory; Basic)

Define behavior and target policies. Explain why Sarsa is on-policy and Q-learning is off-policy by examining both the equations they solve and the samples appearing in their targets. Why does the maximizing action in a Q-learning target not have to be the action actually taken next?

Exercise 7.20. Sarsa and Q-learning on the same transition

(Numerical/derivation; Intermediate)

At s' , action values are 8, 5, -1 . The next action actually sampled has value -1 . For $r = 2$, $\gamma = 0.75$, current value $q(s, a) = 3$, and $\alpha = 0.2$, compute the Sarsa and Q-learning updates. Then repeat if the sampled next action is greedy. Identify the condition under which the two updates coincide.

Exercise 7.21. Identifying policies from a data-generating diagram

(Numerical/derivation; Intermediate)

For each pipeline, identify the behavior policy, target policy, and whether the learning rule is on-policy or off-policy:

- $s_t \xrightarrow{\pi} a_t \rightarrow (r_{t+1}, s_{t+1}) \xrightarrow{\pi} a_{t+1}$, followed by a Sarsa update;
- data generated by a uniform policy β , followed by a Q-learning update and a greedy target policy;
- Q-learning in which the current ε -greedy target policy is also used to choose actions online.

Explain why the last implementation can use the same policy for behavior while the Q-learning rule itself remains off-policy in nature.

Exercise 7.22. Offline Q-learning from a short dataset

(Numerical/derivation; Intermediate)

Let $\gamma = 0.5$, $\alpha = 0.5$, and all initial q-values be zero. Process the fixed dataset in order:

$$(x, a, 2, y), \quad (y, b, 4, x), \quad (x, a, 0, y), \quad (y, a, -1, y).$$

The action set is $\{a, b\}$ at both states. Calculate the full q-table after each transition and the final greedy target policy. Explain which unvisited entry remains uninformed.

Exercise 7.23. Online/offline is different from on/off-policy*(Conceptual/theory; Intermediate)*

Define both pairs of terms and construct the four logically relevant combinations where possible. Why can off-policy Q-learning be implemented online or offline? Why is a strictly on-policy method unable to reuse a fixed dataset generated by a different behavior policy without modification?

Exercise 7.24. Coverage under an off-policy behavior policy*(Numerical/derivation; Challenging)*

At one state with four actions, a behavior policy selects actions with probabilities

$$(0.97, 0.01, 0.01, 0.01).$$

Over $N = 10,000$ visits:

- calculate expected counts for all actions;
- calculate the probability that a particular rare action is never selected;
- repeat the probability for $N = 100$;
- explain why off-policy capability does not remove the requirement that every relevant pair receive enough data.

Part E: A Unified View of TD Targets**Exercise 7.25. Four targets from one trajectory***(Numerical/derivation; Basic)*

Let $r_{t+1} = 1, r_{t+2} = 3, r_{t+3} = -1$, $\gamma = 0.5$, $q(s_{t+1}, a_{t+1}) = 2$, $q(s_{t+2}, a_{t+2}) = 4$, and $\max_a q(s_{t+1}, a) = 5$. Calculate the targets for one-step Sarsa, two-step Sarsa, Q-learning, and the terminating three-reward Monte Carlo case. Use current $q(s_t, a_t) = 0$ and $\alpha = 0.25$ to compute the corresponding four updated values.

Exercise 7.26. Exploration quality limits every sampled method*(Conceptual/theory; Intermediate)*

Explain why an off-policy update formula cannot infer the value of an action that never appears in the data. Contrast a uniform behavior policy with a nearly greedy one in terms of coverage, and distinguish the freedom to choose a behavior policy from freedom to ignore support requirements.

Exercise 7.27. Which equation does each target solve?*(Numerical/derivation; Challenging)*

For Sarsa, n -step Sarsa, Q-learning, and Monte Carlo action-value learning, write the equation $q(s, a) = \mathbb{E}[\bar{q}_t \mid s, a]$ induced by the corresponding target $\bar{q}_t$. Classify each equation as a Bellman equation for a fixed policy or a Bellman optimality equation, and derive the Q-learning classification by taking a maximum over actions.

Exercise 7.28. Capstone: Sarsa and Q-learning along a hazardous route*(Numerical/derivation; Challenging)*

At s' , an ε -greedy behavior policy has three action values 10, 6, -20 and $\varepsilon = 0.3$. A preceding transition from (s, a) gives reward -1 , with $q(s, a) = 4$, $\gamma = 0.9$, and $\alpha = 0.1$.

- Compute the probability of each next action.
- Find the distribution, mean, and variance of the Sarsa target.

- c. Compute the Q-learning target.
- d. Calculate the expected Sarsa update and the Q-learning update.
- e. Interpret the difference as evaluation of exploratory behavior versus direct optimal-value learning.

Exercise 7.29. Error contraction in an expected Q backup

(Numerical/derivation; Challenging)

Define the action-value optimality operator

$$(Tq)(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} q(S_{t+1}, a') \mid s, a].$$

Prove $\|Tq_1 - Tq_2\|_\infty \leq \gamma \|q_1 - q_2\|_\infty$. Explain how a sampled Q-learning target is a noisy observation of this operator and why repeated coverage plus suitable learning rates is necessary in addition to contraction.

Exercise 7.30. One update form, different algorithms

(Conceptual/theory; Challenging)

All the action-value methods can be written $q^+ = q - \alpha(q - \bar{q})$. Explain why sharing this algebraic form does not make the algorithms equivalent. Compare the information, delay, bias, variance, policy relationship, and fixed-point equation encoded by their different targets.

Chapter 8: Value Function Methods

This chapter moves from tabular values to parameterized value functions. It includes linear features, stationary distributions, TD with function approximation, projected Bellman error, least-squares TD, action-value approximation, and deep Q-learning with target networks and experience replay.

Part A: Representation and Features

Exercise 8.1. Fitting a linear value function

(Numerical/derivation; Basic)

Three scalar states have indices $s = 0, 1, 2$ and target values 1, 3, 4. Approximate them by $\hat{v}(s, w) = w_1 s + w_2$.

- Construct the feature matrix Φ and target vector.
- Write the squared-error objective explicitly.
- Solve the normal equations for the least-squares parameter.
- Calculate all fitted values and residuals.
- Explain why three arbitrary values cannot generally be represented exactly with only two parameters.

Exercise 8.2. Storage, accuracy, and generalization

(Conceptual/theory; Basic)

Compare tabular and function representations in how they retrieve a value, update a value, use storage, and generalize across states. Explain why lower-dimensional storage necessarily risks approximation error and why the same parameter sharing that enables generalization can also spread a harmful update.

Exercise 8.3. Polynomial feature construction

(Numerical/derivation; Basic)

For a two-dimensional state $s = (x, y)$, use

$$\phi(s) = (1, x, y, x^2, xy, y^2)^T, \text{quad}w = (1, -2, 3, 0.5, 1, -1)^T.$$

Calculate $\hat{v}(s, w)$ at the four points

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (1, 2),$$

and compute $\nabla_w \hat{v}(s, w)$. Explain why the approximator is linear in w but nonlinear in the state coordinates.

Exercise 8.4. How one parameter update changes many states

(Numerical/derivation; Intermediate)

Let $\hat{v}(s, w) = \phi(s)^T w$, where

$$\phi(s_1) = (1, 0)^T, \quad \phi(s_2) = (1, 1)^T, \quad \phi(s_3) = (1, 2)^T.$$

Starting from $w = (0, 0)^T$, an update based on s_3 changes the parameter by $\Delta w = 0.4\phi(s_3)$. Calculate the old and new estimate at every state. Compare with a tabular update that adds 0.4 only to $v(s_3)$, and quantify the generalization effect.

Exercise 8.5. Choosing features requires structure*(Conceptual/theory; Intermediate)*

Explain why a state index is not necessarily a meaningful numerical coordinate for polynomial fitting. Contrast arbitrary labels with geometric coordinates in a grid world. Discuss the approximation–dimension tradeoff for polynomial and Fourier features and why neural networks are attractive when useful features are unknown.

Part B: Stationary Distributions**Exercise 8.6. Two-step transition probabilities***(Numerical/derivation; Basic)*

For

$$P_\pi = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix},$$

compute P_π^2 and P_π^3 . Interpret entry $(1, 2)$ of each power, and calculate the state distribution after three steps when $d_0 = (1, 0)^T$.

Exercise 8.7. Solving a stationary distribution*(Numerical/derivation; Intermediate)*

Using the matrix in Exercise 8.6, solve $d_\pi^T P_\pi = d_\pi^T$ together with $d_\pi(s_1) + d_\pi(s_2) = 1$. Verify the solution by multiplication. Starting from $d_0 = (0, 1)^T$, compute several successive distributions and compare them with d_π .

Exercise 8.8. Irreducible, regular, and limiting behavior*(Conceptual/theory; Challenging)*

Define accessibility, communication, irreducibility, and regularity. Give a two-state transition matrix that is irreducible but periodic and hence not regular, and explain why its one-time state distributions can oscillate even though a stationary distribution exists. Then add a positive self-transition and explain what changes.

Exercise 8.9. Estimating stationary probabilities from visits*(Numerical/derivation; Intermediate)*

A 10,000-step trajectory visits four states 800, 1200, 2500, 5500 times. Estimate the stationary distribution. For

$$r_\pi = (1, -2, 3, 0)^T, \quad v_\pi = (4, 1, 5, -1)^T,$$

calculate the stationary weighted average immediate reward and average state value. Compare them with uniform-weight averages and interpret the difference.

Part C: TD with State-Value Function Approximation**Exercise 8.10. The weighted approximation objective***(Numerical/derivation; Basic)*

Let true values be $v_\pi = (2, 5, -1)^T$, stationary probabilities $d_\pi = (0.2, 0.3, 0.5)^T$, and estimates $\hat{v} = (w, 2w, w + 1)^T$. Write $J(w) = \sum_s d_\pi(s)(v_\pi(s) - \hat{v}(s, w))^2$, expand it, solve for its minimizer, and compare with the minimizer obtained under uniform state weights.

Exercise 8.11. From gradient descent to a TD semi-gradient update*(Numerical/derivation; Intermediate)*

Starting from $J(w) = \mathbb{E}[(v_\pi(S) - \hat{v}(S, w))^2]$, derive a sample gradient step that would be available if $v_\pi(s_t)$ were known. Replace the unknown value first by a Monte Carlo return and then by $r_{t+1} + \gamma \hat{v}(s_{t+1}, w_t)$. State the resulting two update rules and identify their respective errors.

Exercise 8.12. A linear TD parameter update*(Numerical/derivation; Basic)*

Let $\phi(s_t) = (1, 2)^T$, $\phi(s_{t+1}) = (1, -1)^T$, $w_t = (0.5, 1)^T$, $r_{t+1} = 3$, $\gamma = 0.8$, and $\alpha = 0.1$. Calculate both value estimates, the TD error, the parameter update, and the new estimates at s_t and s_{t+1} .

Exercise 8.13. Why the motivating objective is not the exact TD objective*(Conceptual/theory; Challenging)*

The chapter motivates TD function approximation from squared error to the true value, yet later shows that linear TD minimizes projected Bellman error instead. Explain why replacing the unknown target by a bootstrapped target changes the mathematical problem. Distinguish a motivating derivation from a proof of the actual fixed point.

Exercise 8.14. Fourier feature values*(Numerical/derivation; Intermediate)*

For normalized coordinates $(x, y) = (0.25, 0.5)$ and Fourier order $q = 1$, list the four coefficient pairs $(c_1, c_2) \in \{0, 1\}^2$ and calculate

$$\phi_{c_1, c_2}(s) = \cos(\pi(c_1 x + c_2 y)).$$

Repeat only the newly introduced components when $q = 2$, and verify that the feature dimension is $(q + 1)^2$.

Exercise 8.15. Tabular TD from one-hot features*(Numerical/derivation; Intermediate)*

For n states, take $\phi(s) = e_s$ and $\hat{v}(s, w) = e_s^T w$. Substitute these expressions into the linear TD update and prove that only entry $w(s_t)$ changes. Write the resulting scalar update and identify it with tabular TD learning.

Exercise 8.16. Expected linear TD update in matrix form*(Numerical/derivation; Challenging)*

Let

$$\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.6 \end{bmatrix}, \quad P_\pi = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}, \quad r_\pi = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \gamma = 0.8.$$

Calculate $A = \Phi^T D (I - \gamma P_\pi) \Phi$, $b = \Phi^T D r_\pi$, and $w_* = A^{-1} b$. Verify directly that the expected update $(b - Aw)$ is zero at w_* .

Exercise 8.17. The tabular case as exact representation*(Conceptual/theory; Intermediate)*

Explain why one-hot features with parameter dimension $|S|$ eliminate approximation error but also eliminate the storage advantage. Show qualitatively why $\Phi = I$ makes the TD fixed point equal to v_π , and contrast this with a tall feature matrix having fewer columns than states.

Part D: Projected Bellman Error and Least-Squares TD

Exercise 8.18. Three different error objectives

(Numerical/derivation; Intermediate)

Let $v_\pi = (2, 4)^T$, $\hat{v}(w) = (w, w)^T$, $D = \text{diag}(0.25, 0.75)$, and

$$T_\pi x = r_\pi + \gamma P_\pi x, \text{quad}r_\pi = (1, 0)^T, \text{quad}P_\pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \gamma = 0.5.$$

Compute and minimize the value error $J_E(w)$ and the Bellman error $J_{BE}(w)$. Then construct the D -orthogonal projection onto the span of $(1, 1)^T$, write the projected Bellman equation, and solve its fixed point. Compare the three minimizers.

Exercise 8.19. Deriving the projected Bellman fixed point

(Numerical/derivation; Challenging)

For linear approximation $\hat{v} = \Phi w$, use

$$M = \Phi(\Phi^T D \Phi)^{-1} \Phi^T D$$

and the equation $\Phi w = MT_\pi(\Phi w)$. Derive step by step

$$w = [\Phi^T D(I - \gamma P_\pi) \Phi]^{-1} \Phi^T D r_\pi.$$

Clearly state the full-column-rank condition used to cancel Φ .

Exercise 8.20. Interpreting J_E , J_{BE} , and J_{PBE}

(Conceptual/theory; Challenging)

For each objective, state what discrepancy it measures, whether it directly requires the unknown true value, and whether linear TD minimizes it. Explain geometrically why projection is necessary when the Bellman image falls outside the approximation subspace.

Exercise 8.21. Using the approximation error bound

(Numerical/derivation; Intermediate)

Suppose the best representable value has $\min_w \|\hat{v}(w) - v_\pi\|_D = 0.08$. Use the chapter's bound to estimate an upper bound on the TD fixed-point error for $\gamma = 0.5, 0.9, 0.99$. Discuss why the bound becomes weak near one and why a loose bound is not a claim that the actual approximation must be poor.

Exercise 8.22. Constructing LSTD estimates

(Numerical/derivation; Intermediate)

For $\gamma = 0.5$, a trajectory provides

t	$\phi(s_t)$	$(r_{t+1}, \phi(s_{t+1}))$
0	$(1, 0)^T$	$(2, (1, 1)^T)$
1	$(1, 1)^T$	$(-1, (0, 1)^T)$
2	$(0, 1)^T$	$(3, (1, 0)^T)$

Calculate $\hat{A} = \sum_t \phi(s_t)(\phi(s_t) - \gamma\phi(s_{t+1}))^T$, $\hat{b} = \sum_t r_{t+1}\phi(s_t)$, and, if invertible, $\hat{w} = \hat{A}^{-1}\hat{b}$. If an intermediate $\hat{A}$ is singular, show how adding σI with $\sigma = 0.1$ changes it.

Exercise 8.23. A recursive rank-one LSTD update*(Numerical/derivation; Challenging)*

Show that adding one sample changes $\hat{A}_t$ by the rank-one matrix $u_t v_t^T = \phi(s_t)(\phi(s_t) - \gamma\phi(s_{t+1}))^T$. Using the rank-one inverse identity

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u},$$

calculate $\hat{A}_{t+1}^{-1}$ for

$$\hat{A}_t^{-1} = I_2, \quad u_t = (1, 2)^T, \quad v_t = (0.5, -0.5)^T,$$

provided the denominator is nonzero. Verify by multiplying against $\hat{A}_t + u_t v_t^T$.

Exercise 8.24. When LSTD is attractive*(Conceptual/theory; Intermediate)*

Compare LSTD and incremental linear TD in sample efficiency, per-update computation, storage, need for matrix inversion or inverse maintenance, support for nonlinear approximators, and extension to action-value control. Explain why faster convergence in samples need not mean lower computational cost.

Part E: Action Values and Deep Q-Learning**Exercise 8.25. Sarsa with action-value approximation***(Numerical/derivation; Basic)*

Let $\hat{q}(s_t, a_t, w_t) = 2$, $\hat{q}(s_{t+1}, a_{t+1}, w_t) = 5$, $\nabla_w \hat{q}(s_t, a_t, w_t) = (1, -1)^T$, $r_{t+1} = -1$, $\gamma = 0.8$, $\alpha = 0.1$, and $w_t = (2, 3)^T$. Calculate the TD error and the new parameter. Explain why the gradient is taken at the current pair rather than at the next pair.

Exercise 8.26. Q-learning with linear features*(Numerical/derivation; Intermediate)*

At s_t, a_t , $\phi = (1, 2)^T$ and $w = (1, -0.5)^T$. At s_{t+1} , the feature vectors for actions a, b, c are $(1, 0)^T, (1, 1)^T, (1, -1)^T$. Let $r = 2$, $\gamma = 0.9$, and $\alpha = 0.05$. Compute every next action estimate, the Q-learning target, TD error, parameter update, and new estimate of the visited pair.

Exercise 8.27. A value function does not remove a tabular policy*(Conceptual/theory; Intermediate)*

In the chapter's Sarsa and Q-learning function-approximation algorithms, values are functions but an ε -greedy policy may still be stored in a table. Explain why this retains a finite-state/action limitation and how the next chapter's policy-function idea addresses it. Do not assume any specific policy-gradient formula.

Exercise 8.28. One deep Q-learning mini-batch*(Numerical/derivation; Intermediate)*

A mini-batch contains two transitions. The main-network predictions for the visited pairs are 2.0, 4.0; the rewards are 1, -2; and the target-network maximum next-state values are 5, 3. Let $\gamma = 0.8$. Calculate both target values, both TD residuals, and the mean squared loss. Then calculate $\partial L / \partial y_i$ for each main-network output y_i , treating the target values as fixed.

Exercise 8.29. Experience replay changes the sample distribution*(Numerical/derivation; Challenging)*

A replay buffer has 1000 transitions: 700 from region A , 200 from B , and 100 from C . A mini-batch of size 50 is sampled uniformly from buffer entries.

- a. Find the expected number from each region.
- b. Find the probability that no C transition appears if sampling is with replacement.
- c. Compare this with processing the most recent 50 sequential samples if all of them happen to be from A .
- d. Explain how replay breaks temporal correlation and permits repeated use without manufacturing coverage absent from the buffer.

Exercise 8.30. Training loss versus value accuracy

(Conceptual/theory; Challenging)

A deep Q-network attains nearly zero replay-buffer loss, but its true action values remain inaccurate on many state–action pairs. Explain how this can happen when the buffer is small or unrepresentative. Distinguish fitting the available targets, satisfying the Bellman optimality equation broadly, and generalizing to unseen pairs. Include the separate role of the periodically updated target network.

Chapter 9: Policy Gradient Methods

This chapter covers policy-function representation, scalar performance metrics, the policy gradient theorem in discounted and undiscounted settings, softmax policies, the Poisson equation, and Monte Carlo policy gradient (REINFORCE). Baselines and actor-critic extensions are reserved for Chapter 10.

Part A: Policies as Parameterized Functions

Exercise 9.1. Softmax action probabilities

(Numerical/derivation; Basic)

At a state s , a policy has preferences $h(s, a_1, \theta) = 1$, $h(s, a_2, \theta) = 0$, and $h(s, a_3, \theta) = -1$. Using

$$\pi(a \mid s, \theta) = \frac{e^{h(s, a, \theta)}}{\sum_{a'} e^{h(s, a', \theta)}},$$

calculate all action probabilities to four decimal places. Verify that they sum to one, identify the most probable action, and show that adding the same constant to all preferences leaves the probabilities unchanged.

Exercise 9.2. Table versus policy function

(Conceptual/theory; Basic)

Compare tabular and function policy representations in how action probabilities are stored, retrieved, and updated. Explain the storage and generalization benefits of parameter sharing and why changing one parameter can alter the policy at many states simultaneously.

Exercise 9.3. A logistic two-action policy

(Numerical/derivation; Basic)

Let

$$\pi(a_1 \mid s, \theta) = \frac{e^{\theta x_s}}{1 + e^{\theta x_s}}, \quad \pi(a_2 \mid s, \theta) = \frac{1}{1 + e^{\theta x_s}}.$$

- Calculate both probabilities for $x_s = 2, \theta = 0.4$.
- Derive $\partial \pi(a_i \mid s, \theta) / \partial \theta$ for both actions.
- Derive both score functions $\partial \ln \pi(a_i \mid s, \theta) / \partial \theta$.
- Verify that the policy-weighted mean of the two score functions is zero.

Exercise 9.4. A direct gradient-ascent policy update

(Numerical/derivation; Intermediate)

A scalar objective has gradient $\nabla_{\theta} J(\theta) = 2 - 0.5\theta$. Starting from $\theta_0 = -2$ with step size $\alpha = 0.4$, perform five gradient-ascent updates. Solve for the stationary point analytically and show how the error evolves. Explain why this calculation represents the optimization stage of a policy-gradient method but not yet the derivation of its reinforcement-learning gradient.

Exercise 9.5. Optimality through a scalar metric

(Conceptual/theory; Intermediate)

Earlier tabular chapters defined an optimal policy by componentwise dominance of all state values. Explain why a parameterized policy is instead optimized through a chosen scalar metric. Can two metrics prefer different policies if they weight starting states differently? Give a qualitative two-state example.

Part B: Performance Metrics

Exercise 9.6. Average state value under different weights

(Numerical/derivation; Basic)

A policy has state values $v_\pi = (8, 2, -1, 5)^T$. Calculate $\bar{v}_\pi = d^T v_\pi$ for:

- the uniform distribution;
- $d_0 = (1, 0, 0, 0)^T$;
- $d = (0.1, 0.2, 0.6, 0.1)^T$.

Construct a second value vector whose ordering relative to v_π changes between at least two of these distributions.

Exercise 9.7. Discounted reward sum equals an initial average value

(Numerical/derivation; Intermediate)

The initial state is x with probability 0.3 and y with probability 0.7. Under a policy, the deterministic reward sequences are

$$x : (2, 2, 2, \dots), \quad y : (-1, 4, 0, 0, \dots),$$

with $\gamma = 0.5$. Calculate $v_\pi(x), v_\pi(y)$, their initial weighted average, and the expectation of the full discounted reward sum. Verify the two expressions agree.

Exercise 9.8. Policy-independent and stationary state weights

(Conceptual/theory; Intermediate)

Contrast d_0 , which is fixed independently of the policy, with d_π , which is the stationary distribution induced by the policy. When is each a natural choice? Explain why differentiating $d_0^T v_\pi$ is simpler than differentiating $d_\pi^T v_\pi$.

Exercise 9.9. Average reward from a stationary distribution

(Numerical/derivation; Basic)

Let

$$d_\pi = (0.2, 0.5, 0.3)^T, \quad r_\pi = (3, -1, 2)^T.$$

Calculate $\bar{r}_\pi$. A 10,000-step run visits the states 1950, 5100, 2950 times and observes mean one-step rewards 3.1, -0.9, 1.8 within them. Estimate the long-run average reward from these observations and compare it with the theoretical value.

Exercise 9.10. Average value and average reward are not always interchangeable

(Conceptual/theory; Challenging)

State the discounted relation $\bar{r}_\pi = (1 - \gamma)\bar{v}_\pi$ when stationary weights are used. Explain why it implies the same maximizers for fixed γ , and why it does not justify replacing an arbitrary fixed-initial-distribution metric $d_0^T v_\pi$ by a stationary average reward without further analysis.

Part C: The Policy Gradient in the Discounted Case

Exercise 9.11. Discounted total transition probabilities

(Numerical/derivation; Intermediate)

For

$$P_\pi = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}, \quad \gamma = 0.5,$$

calculate $(I - \gamma P_\pi)^{-1}$. Verify the result by multiplication and interpret entry $(1, 2)$ as $\sum_{k=0}^{\infty} \gamma^k [P_\pi^k]_{12}$. Approximate that entry using the first four powers and compare with the exact value.

Exercise 9.12. Constructing the discounted state-weight measure

(Numerical/derivation; Intermediate)

Using the matrix from Exercise 9.11 and initial distribution $d_0 = (0.6, 0.4)^T$, compute

$$\rho_\pi^T = d_0^T (I - \gamma P_\pi)^{-1}.$$

Calculate the sum of the entries of ρ_π and explain why it is not generally one under the chapter's unnormalized discounted-total-probability definition. Normalize it and compare the two ways of weighting a policy gradient.

Exercise 9.13. Evaluating a finite policy gradient

(Numerical/derivation; Basic)

At two states, the state weights are $\eta = (0.4, 0.6)^T$. There are two actions, and a scalar parameter has policy derivatives and action values

	$\partial_\theta \pi(a_1 s)$	$\partial_\theta \pi(a_2 s)$	$q_\pi(s, a_1)$	$q_\pi(s, a_2)$
s_1	0.2	-0.2	5	1
s_2	-0.1	0.1	2	6

Calculate $\sum_s \eta(s) \sum_a \partial_\theta \pi(a | s) q_\pi(s, a)$. Identify which state pushes the parameter upward and which pushes it downward.

Exercise 9.14. The log-derivative trick and positive policies

(Conceptual/theory; Basic)

Derive $\nabla_\theta \pi = \pi \nabla_\theta \ln \pi$ and explain why it converts a sum over actions into an expectation under the policy. Why must action probabilities be positive for the logarithm? Explain how a softmax output meets positivity and normalization simultaneously.

Exercise 9.15. Score gradient of a softmax policy

(Numerical/derivation; Intermediate)

Let preferences be $h_i = \theta_i$ for three actions and $\pi_i = e^{\theta_i} / \sum_j e^{\theta_j}$.

- Derive $\partial \ln \pi_i / \partial \theta_j = \mathbf{1}\{i = j\} - \pi_j$.
- Evaluate all three score-gradient vectors at equal preferences.
- For action values $(6, 0, 3)$, calculate the expected policy-gradient vector under the uniform policy.
- Verify that its components sum to zero and interpret the signs.

Exercise 9.16. Unrolling the gradient of a state value

(Numerical/derivation; Challenging)

Starting from $v_\pi(s) = \sum_a \pi(a | s, \theta) q_\pi(s, a)$, differentiate with respect to θ . Substitute $q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} p(s' | s, a) v_\pi(s')$, and derive a recursive equation for $\nabla_\theta v_\pi$. Convert it to matrix form and solve it using $(I - \gamma P_\pi)^{-1}$, recovering the discounted total transition weights.

Exercise 9.17. A complete two-state policy-gradient calculation

(Numerical/derivation; Challenging)

Let

$$P_\pi = \begin{bmatrix} 0.5 & 0.5 \\ 0.25 & 0.75 \end{bmatrix}, \quad d_0 = (1, 0)^T, \quad \gamma = 0.8.$$

At s_1 , $\sum_a \nabla_{\theta} \pi(a | s_1) q_{\pi}(s_1, a) = (2, -1)^T$; at s_2 , it is $(-1, 3)^T$. Calculate ρ_{π} and then the two-dimensional gradient of $\bar{v}_{\pi}^0$. Show all matrix and weighted-sum steps.

Exercise 9.18. Using the stationary policy-gradient approximation

(Numerical/derivation; Intermediate)

Suppose $d_{\pi} = (0.25, 0.75)^T$, $\gamma = 0.95$, and the per-state gradient contributions $u(s) = \sum_a \nabla_{\theta} \pi(a | s) q_{\pi}(s, a)$ are $u(s_1) = (2, 0)^T$, $u(s_2) = (-1, 4)^T$. Calculate the stationary approximation to $\nabla_{\theta} \bar{r}_{\pi}$ and to $\nabla_{\theta} \bar{v}_{\pi}$. Explain the factor $1/(1 - \gamma)$.

Exercise 9.19. Verifying $\bar{r}_{\pi} = (1 - \gamma)\bar{v}_{\pi}$

(Numerical/derivation; Intermediate)

Starting from $v_{\pi} = r_{\pi} + \gamma P_{\pi} v_{\pi}$ and $d_{\pi}^T P_{\pi} = d_{\pi}^T$, derive the metric identity. Verify it numerically for

$$P_{\pi} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}, \quad r_{\pi} = (1, 3)^T, \quad \gamma = 0.5,$$

by solving the Bellman equation and stationary distribution.

Exercise 9.20. Why one discounted gradient is approximate

(Conceptual/theory; Challenging)

When differentiating $d_{\pi}^T v_{\pi}$, two terms appear because both the stationary distribution and the value depend on the policy. Explain which term is neglected in the chapter's approximation and why the retained term becomes dominant as $\gamma \rightarrow 1$. Contrast this approximation with the exact fixed-initial-distribution theorem.

Part D: The Undiscounted Case and the Poisson Equation

Exercise 9.21. Differential returns

(Numerical/derivation; Basic)

An undiscounted continuing policy has average reward $\bar{r}_{\pi} = 2$. From state s , a deterministic reward sequence is $5, 1, 2, 2, 2, \dots$. Calculate the differential return $\sum_{k \geq 0} (R_{t+k+1} - \bar{r}_{\pi})$. Compare it with the ordinary undiscounted sum of rewards and explain why centering by the average makes the value finite here.

Exercise 9.22. Solving a Poisson equation

(Numerical/derivation; Challenging)

Let

$$P_{\pi} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}, \quad r_{\pi} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

- Find d_{π} and $\bar{r}_{\pi}$.
- Write the Poisson equation $v_{\pi} = r_{\pi} - \bar{r}_{\pi} \mathbf{1} + P_{\pi} v_{\pi}$.
- Solve it with the additional normalization $d_{\pi}^T v_{\pi} = 0$.
- Verify that adding any constant multiple of $\mathbf{1}$ gives another solution.
- Compute the particular solution $(I - P_{\pi} + \mathbf{1} d_{\pi}^T)^{-1} r_{\pi}$ and relate it to yours.

Exercise 9.23. Why undiscounted differential values are nonunique*(Conceptual/theory; Intermediate)*

Explain algebraically why $I - P_\pi$ is singular and why adding $c\mathbf{1}$ to a solution of the Poisson equation changes no relative values. Why is the average reward nevertheless unique? Give one possible normalization that selects a unique representative.

Exercise 9.24. Undiscounted average-reward gradient*(Numerical/derivation; Challenging)*

Starting from the differentiated Poisson equation, derive the cancellation that occurs after multiplying by d_π^T , and show

$$\nabla_\theta \bar{r}_\pi = \sum_s d_\pi(s) \sum_a \nabla_\theta \pi(a | s, \theta) q_\pi(s, a).$$

Your derivation must explicitly use $d_\pi^T P_\pi = d_\pi^T$ and $d_\pi^T \mathbf{1} = 1$.

Exercise 9.25. Discounted and undiscounted action values*(Numerical/derivation; Intermediate)*

A continuing reward sequence from a fixed state–action pair is $4, 0, 2, 2, 2, \dots$, and the policy’s long-run average reward is 2.

- Calculate the discounted return at $\gamma = 0.8$.
- Calculate the differential undiscounted return.
- Explain what different aspect of performance each value measures.
- Analyze the discounted return as $\gamma \uparrow 1$ and compare it with the finite differential value.

Part E: Monte Carlo Policy Gradient (REINFORCE)**Exercise 9.26. Returns and REINFORCE updates from one episode***(Numerical/derivation; Basic)*

An episode has rewards $2, -1, 4$, then terminates, with $\gamma = 0.5$. The scalar score gradients at its three state–action pairs are $0.4, -0.2, 0.5$. Starting from $\theta = 1$ with $\alpha = 0.1$:

- compute every return by working backward;
- process the three REINFORCE updates in chronological order, treating the listed score gradients as fixed numerical inputs;
- identify which visits increase and which decrease the parameter.

Exercise 9.27. Why basic REINFORCE is on-policy*(Conceptual/theory; Intermediate)*

The policy-gradient expectation samples actions from the policy being optimized. Explain why an episode generated by another policy does not directly give the required stochastic gradient. Also discuss the state-sample distribution and why using every visit in one episode is more sample-efficient than drawing a fresh episode for each individual update, even though it is not the ideal independent sampling scheme.

Exercise 9.28. Interpreting the sign and scale of a policy update*(Numerical/derivation; Intermediate)*

For a scalar parameter, write the REINFORCE step as

$$\theta^+ = \theta + \alpha \beta \nabla_{\theta} \pi(a \mid s, \theta), \quad \beta = \frac{q(s, a)}{\pi(a \mid s, \theta)}.$$

For the four cases $(q, \pi) = (6, 0.2), (6, 0.8), (-3, 0.2), (-3, 0.8)$, calculate β . Assuming the same positive scalar policy derivative, rank the magnitude and direction of the probability changes and interpret exploitation and exploration effects.

Exercise 9.29. Capstone: one softmax REINFORCE step*(Numerical/derivation; Challenging)*

At one state, a three-action softmax policy is initially uniform. A sampled episode selects a_2 there and produces return $G = 5$. Let the preference parameters be $\theta = (0, 0, 0)^T$ and $\alpha = 0.06$.

- Compute $\nabla_{\theta} \ln \pi(a_2 \mid s, \theta)$.
- Perform the REINFORCE parameter update.
- Calculate the new softmax probabilities.
- Repeat for return $G = -5$.
- Explain why increasing one preference changes all action probabilities.

Exercise 9.30. Strengths and limits of Monte Carlo policy gradient*(Conceptual/theory; Challenging)*

Explain how REINFORCE combines a parameterized stochastic policy, a Monte Carlo action-value estimate, and stochastic gradient ascent. Discuss its ability to handle policy functions and its costs: episodic waiting, return variance, and on-policy sampling. Identify, without giving formulas from the next chapter, the two broad ideas that actor–critic methods add to address value estimation and variance.

Chapter 10: Actor–Critic Methods

This chapter covers Q actor–critic, advantage actor–critic, baseline invariance and variance reduction, importance sampling for off-policy actor–critic, and deterministic policy gradients.

Part A: The Basic Actor–Critic Structure

Exercise 10.1. One Q actor–critic step

(Numerical/derivation; Basic)

At time t , suppose

$$q(s_t, a_t, w_t) = 3, quadq(s_{t+1}, a_{t+1}, w_t) = 5, quadr_{t+1} = -1,$$

$\gamma = 0.8$, $\nabla_w q(s_t, a_t, w_t) = (1, 2)^T$, $\nabla_\theta \ln \pi(a_t \mid s_t, \theta_t) = (-0.5, 1)^T$, $w_t = (0, 1)^T$, $\theta_t = (2, -1)^T$, $\alpha_w = 0.1$, and $\alpha_\theta = 0.05$. Calculate the critic TD error, new critic parameter, and actor parameter update using the current action-value estimate.

Exercise 10.2. Why the names actor and critic fit

(Conceptual/theory; Basic)

Explain which component chooses actions and which evaluates them. Why is an actor–critic method still a policy-gradient method? Why is the combined structure simultaneously policy based and value based?

Exercise 10.3. Monte Carlo critic versus TD critic

(Numerical/derivation; Intermediate)

For a visited pair, the current estimate is $q = 2$. A complete return is $G = 7$, while the one-step sample has $r = 1$, next estimate $q' = 4$, and $\gamma = 0.9$. With $\alpha_w = 0.2$ and unit value gradient, calculate the critic update using the Monte Carlo target and using the TD target. Then calculate the actor coefficient supplied by each updated estimate and discuss the numerical difference.

Exercise 10.4. A two-step QAC trace

(Numerical/derivation; Intermediate)

Use scalar linear functions $q(s, a, w) = w\phi(s, a)$ and score values $z_t = \partial_\theta \ln \pi(a_t \mid s_t, \theta)$. Let $w_0 = 1, \theta_0 = 0$, $\alpha_w = 0.1$, $\alpha_\theta = 0.05$, $\gamma = 0.5$, and two samples have

t	ϕ_t	r_{t+1}	ϕ_{t+1}	z_t
0	2	1	1	0.4
1	1	-1	3	-0.2

Process actor then critic at each time using the current parameters, and report all intermediate q-values, TD errors, and parameters.

Exercise 10.5. REINFORCE and QAC differ only in value estimation

(Conceptual/theory; Intermediate)

Explain the common policy-gradient update shared by REINFORCE and QAC. What changes when the action value is estimated from a full return versus a TD critic? Compare update timing, bootstrapping, and expected variance without claiming that one is universally superior.

Part B: Baselines and Advantage Actor–Critic

Exercise 10.6. Proving baseline invariance

(Numerical/derivation; Intermediate)

Let $b(s)$ depend on the state but not on the action. Prove

$$\mathbb{E}_{A \sim \pi(\cdot | s)}[\nabla_{\theta} \ln \pi(A | s, \theta) b(s)] = 0.$$

Expand the expectation as a sum, use $\pi \nabla \ln \pi = \nabla \pi$, and use normalization of the policy. Extend the proof to an outer expectation over states.

Exercise 10.7. Computing the variance-minimizing baseline

(Numerical/derivation; Challenging)

At one state, a scalar score $z(a) = \nabla_{\theta} \ln \pi(a | s)$, policy probability, and action value are

	a_1	a_2	a_3
$\pi(a s)$	0.5	0.3	0.2
$z(a)$	1	−2	0.5
$q_{\pi}(s, a)$	6	1	4

Calculate the optimal baseline

$$b_*(s) = \frac{\mathbb{E}[z(A)^2 q_{\pi}(s, A)]}{\mathbb{E}[z(A)^2]},$$

the simpler state-value baseline $v_{\pi}(s)$, and zero. For each baseline, calculate the mean and variance of the scalar gradient sample $z(A)(q_{\pi}(s, A) - b)$, and compare them.

Exercise 10.8. Why the baseline must not depend on the sampled action

(Conceptual/theory; Challenging)

The standard invariance proof permits $b(S)$, not an arbitrary $b(S, A)$. Identify the exact algebraic step that fails for an action-dependent baseline. Construct a two-action numerical example where subtracting $b(s, a) = q(s, a)$ changes the expected gradient to zero even though the original gradient is nonzero.

Exercise 10.9. Advantages are centered action values

(Numerical/derivation; Basic)

At a state, action probabilities are (0.5, 0.3, 0.2) and action values are (8, 2, −1).

- Calculate $v_{\pi}(s)$.
- Calculate every advantage $q_{\pi}(s, a) - v_{\pi}(s)$.
- Verify that the policy-weighted average advantage is zero.
- Identify which sampled actions would reinforce and which would weaken their log-probability directions when score signs are held fixed.

Exercise 10.10. One A2C actor and critic update

(Numerical/derivation; Basic)

Let

$$v(s_t, w_t) = 2, \text{quadv}(s_{t+1}, w_t) = 4, \text{quadr}_{t+1} = 1, \text{quad}\gamma = 0.75,$$

$\nabla_w v(s_t, w_t) = (1, -1)^T$, $\nabla_\theta \ln \pi(a_t | s_t, \theta_t) = (0.2, 0.6)^T$, $w_t = (1, 2)^T$, $\theta_t = (0, 0)^T$, and $\alpha_w = 0.1, \alpha_\theta = 0.05$. Calculate the TD-error advantage and both parameter updates.

Exercise 10.11. Quantifying variance reduction by centering

(Numerical/derivation; Intermediate)

Under a fixed state and policy, a scalar score is $Z \in \{-1, +1\}$ with equal probability, and the paired action values are $Q = 9$ when $Z = +1$ and $Q = 7$ when $Z = -1$.

- Calculate the mean and variance of (ZQ) with zero baseline.
- Use baseline $b = 8$, then calculate the mean and variance of $Z(Q - b)$.
- Verify that the mean is invariant and interpret the variance change.

Exercise 10.12. Why the TD error can stand in for an advantage

(Conceptual/theory; Intermediate)

Starting from the definition of $q_\pi(s, a)$, explain why $R_{t+1} + \gamma v_\pi(S_{t+1}) - v_\pi(s)$ is a sample whose conditional expectation is the advantage. Why does using it require only a state-value critic rather than separate state- and action-value networks? Distinguish a sample advantage from the exact advantage.

Part C: Importance Sampling and Off-Policy Actor–Critic

Exercise 10.13. A basic importance-sampling estimate

(Numerical/derivation; Basic)

Let $X \in \{-2, 1, 4\}$. The target distribution is $p_0 = (0.2, 0.5, 0.3)$, and the sampling distribution is $p_1 = (0.5, 0.25, 0.25)$. A sample from p_1 is $(-2, -2, 1, 4, -2, 1, 4, 4)$.

- Calculate the ordinary sample mean.
- Calculate every possible importance weight $p_0(x)/p_1(x)$.
- Calculate the importance-weighted estimate of $\mathbb{E}_{p_0}[X]$.
- Compare both estimates with the exact target expectation.

Exercise 10.14. Expectation and variance of an importance estimator

(Numerical/derivation; Challenging)

For a single sample $X \sim p_1$, define $Y = [p_0(X)/p_1(X)]X$ using the distributions in Exercise 10.13. Calculate $\mathbb{E}_{p_1}[Y]$, $\mathbb{E}_{p_1}[Y^2]$, and $\text{var}_{p_1}(Y)$. Compare with the variance of a direct sample $X \sim p_0$. Then derive the variance of the mean of n independent importance-weighted samples.

Exercise 10.15. The support condition

(Conceptual/theory; Basic)

Explain why the behavior distribution must assign positive probability wherever the target distribution does. Give a three-outcome example in which an outcome required by the target is impossible under the behavior policy and show why no amount of behavior data can repair the missing contribution.

Exercise 10.16. Importance weights for policy actions

(Numerical/derivation; Basic)

At a state, the target policy is $\pi = (0.6, 0.3, 0.1)$ and the behavior policy is $\beta = (0.2, 0.5, 0.3)$.

- Calculate the importance weight for each action.

- b. Verify that $\mathbb{E}_{A \sim \beta}[\pi(A)/\beta(A)] = 1$.
- c. For action-dependent quantities $f = (4, -1, 2)$, verify numerically that the behavior expectation of the weighted quantity equals $\mathbb{E}_{A \sim \pi}[f(A)]$.

Exercise 10.17. An off-policy actor update

(Numerical/derivation; Intermediate)

A behavior policy selects the observed action with probability 0.1, while the target policy assigns it probability 0.4. The TD error is -2 , the score gradient is $(0.5, -1)^T$, $\alpha_\theta = 0.03$, and $\theta = (1, 2)^T$. Calculate the importance weight and new actor parameter. Repeat if the behavior probability is 0.8, and compare update magnitudes.

Exercise 10.18. An off-policy critic update

(Numerical/derivation; Intermediate)

For the same probability ratios as Exercise 10.17, suppose $\nabla_w v(s, w) = (2, 1)^T$, $w = (0, -1)^T$, $\alpha_w = 0.05$, and TD error -2 . Calculate both critic updates. Explain why the chapter applies the importance weight to the value update as well as the policy update.

Exercise 10.19. Baseline invariance with importance weights

(Numerical/derivation; Challenging)

Prove

$$\mathbb{E}_{A \sim \beta} \left[\frac{\pi(A | s)}{\beta(A | s)} \nabla_\theta \ln \pi(A | s) b(s) \right] = 0.$$

State the support assumption needed to manipulate the ratio. Then show how the result yields an off-policy advantage-gradient expression.

Exercise 10.20. Capstone: on-policy and off-policy gradient estimates

(Numerical/derivation; Challenging)

At one state, two actions have target probabilities $(0.75, 0.25)$, behavior probabilities $(0.4, 0.6)$, scalar score gradients $(0.25, -0.75)$, and advantages $(2, -6)$.

- a. Calculate the exact target-policy expected gradient.
- b. Calculate the importance-weighted behavior-policy expected gradient and verify equality.
- c. List the two possible one-sample off-policy estimates and their probabilities.
- d. Calculate their variance and compare with direct on-policy sampling.
- e. Explain why unbiased correction can still have high variance.

Exercise 10.21. Why both actor and critic need off-policy care

(Conceptual/theory; Challenging)

Explain separately how policy mismatch affects the actor's action distribution and the critic's evaluation target. Why is correcting only the actor generally insufficient in the chapter's off-policy A2C construction? Discuss the danger of very large importance weights when the behavior gives an observed action a tiny probability.

Part D: Deterministic Actor–Critic

Exercise 10.22. A scalar deterministic policy gradient

(Numerical/derivation; Basic)

Let the deterministic policy be $\mu(s, \theta) = \theta s$ and the critic be $q(s, a) = -(a - 2s)^2 + 5$. For a state sample $s = 3$:

- calculate $\partial\mu/\partial\theta$;
- calculate $\partial q/\partial a$ at $a = \mu(s, \theta)$;
- calculate the deterministic policy-gradient sample at $\theta = 1$;
- perform one ascent update with $\alpha_\theta = 0.02$;
- identify the parameter value that makes the policy choose the critic-maximizing action for every nonzero s .

Exercise 10.23. Why deterministic policy gradients are naturally off-policy

(Conceptual/theory; Intermediate)

Contrast the random-action expectation in the stochastic policy gradient with the state-only expectation in the deterministic theorem. Explain why behavior actions can come from another exploratory policy while the actor gradient is evaluated at $a = \mu(s)$. Clarify why state coverage is still required.

Exercise 10.24. Chain rule through actor and critic

(Numerical/derivation; Intermediate)

Let a two-dimensional actor output

$$\mu(s, \theta) = \begin{bmatrix} \theta_1 s \\ \theta_2 s^2 \end{bmatrix},$$

and let $q(s, a_1, a_2) = 3a_1 - a_1^2 + 2a_2 - \frac{1}{2}a_2^2$. At $s = 2$, $\theta = (0.5, -0.25)^T$, compute the actor Jacobian, $\nabla_a q$, and the full parameter-gradient sample using the chain rule.

Exercise 10.25. Discounted state weights for a deterministic policy

(Numerical/derivation; Intermediate)

Under a deterministic policy,

$$P_\mu = \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}, \quad d_0 = (1, 0)^T, \quad \gamma = 0.5.$$

Calculate $\rho_\mu^T = d_0^T (I - \gamma P_\mu)^{-1}$. If the per-state deterministic gradient contributions are $u(s_1) = 2$ and $u(s_2) = -1$, calculate the total gradient of the initial-distribution objective.

Exercise 10.26. One deterministic actor-critic update

(Numerical/derivation; Intermediate)

Let

$$\mu(s, \theta) = \theta s, \quad q(s, a, w) = w_1 s + w_2 a,$$

with $s_t = 2, a_t = 1, r_{t+1} = 3, s_{t+1} = 1, \theta = 0.5, w = (1, 2)^T, \gamma = 0.8, \alpha_\theta = 0.05, \alpha_w = 0.1$. Calculate the next target action $\mu(s_{t+1})$, TD error, actor gradient, new actor parameter, critic gradient at the behavior pair, and new critic parameter.

Exercise 10.27. Why deterministic policies help with continuous actions

(Conceptual/theory; Basic)

Explain why a deterministic actor can output a continuous action directly, whereas a finite-action softmax does not naturally do so. Why must the critic be differentiable with respect to its action input? Explain the need for a separate exploratory behavior policy or added action noise during data collection.

Exercise 10.28. Deriving the deterministic value gradient recursion*(Numerical/derivation; Challenging)*

Starting from $v_\mu(s) = q_\mu(s, \mu(s))$, apply the chain rule and the action-value Bellman equation to derive

$$\nabla_\theta v_\mu(s) = \nabla_\theta \mu(s) \nabla_a q_\mu(s, a)|_{a=\mu(s)} + \gamma \sum_{s'} p(s' | s, \mu(s)) \nabla_\theta v_\mu(s').$$

Solve the coupled equations with a discounted transition resolvent and recover the deterministic policy-gradient theorem for a fixed initial distribution.

Exercise 10.29. Capstone: noisy behavior and a deterministic target*(Numerical/derivation; Challenging)*

The deterministic target policy is $\mu(s, \theta) = \theta s$, but the behavior action is $a_t = \mu(s_t, \theta) + \xi_t$. Let $q(s, a, w) = -(a - ws)^2 + 4s$, with $s_t = 2$, $s_{t+1} = 1$, $r_{t+1} = 1$, $\theta = 1$, $w = 0.5$, behavior noise $\xi_t = 1$, $\gamma = 0.9$, and unit scalar learning rates left symbolically as α_θ, α_w .

- Calculate the behavior action and target next action.
- Calculate the critic TD error.
- Derive the actor-gradient sample at the target action, not the behavior action.
- Derive the critic-gradient sample at the behavior pair.
- Write both parameter updates and explain the two different action arguments.

Exercise 10.30. Comparing the actor–critic family*(Conceptual/theory; Challenging)*

Compare QAC, A2C, off-policy A2C with importance sampling, and deterministic actor–critic along these dimensions: critic quantity, actor signal, baseline, behavior versus target policy, exploration mechanism, and suitability for continuous actions. Explain the common actor–critic structure that remains unchanged across all four.